Decision Analysis

Manuscript In Progress

Ronald A. Howard

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1 Possibilities and Probabilities

Ronald A. Howard

Section 1: Possibilities and Probabilities

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1.1 Possibilities

To define a decision, we must have a precise way of characterizing the possible consequen of the decision. We call this "creating the possibilities." Since any possibility can be represe in terms of events, we begin by characterizing events.

1.1.1 Events and Distinctions

An event is a distinction and the simplest event is a simple distinction; that is, a division of all eventualities into two states. For example, suppose that we are sitting in a public room, a meeting room or a lecture hall, and considering whether the next person who enters the room will be a beer drinker. Once we have defined what we mean by beer drinker, then the next person entering will either be a beer drinker or will not be a beer drinker. There are only two possibilities.

What we mean by beer drinker in any setting depends on the background of understanding of the person involved. Often the distinction will be sufficiently clear that no more specification of it is necessary. However, the everyday meaning of terms contains sufficient ambiguity that it is not adequate for the careful analysis of decisions. For example, by beer drinker, do we mean someone who has ever drunk beer? Someone who drinks beer regularly? Someone who drinks mainly beer as an alcoholic beverage? The more we examine the distinction, the less clear it becomes.

1.1.2 The Clarity Test

To ensure that we are dealing only with clear distinctions in an analysis, we must require that every event meet the clarity test. The clarity test rests on the notion of a clairvoyant, which we have previously defined. As you recall, a clairvoyant is both competent and trustworthy. He knows the value of any physically defined quantity, both in the future and in the past. In addition, the clairvoyant can perform any amount of computation or unit conversion required. If the event is sufficiently specified that a clairvoyant can tell without any exercise of judgment whether that event has occurred, then the event specification passes the clarity test.

Now let us examine the event "beer drinker." How can we define what we mean by Beer Drinker, where we capitalize the words to show that we are using the term in a particular sense? We know that a Beer Drinker is a human being who drinks a certain quantity of beer. We might specify consumption of at least one hundred quarts of beer per year. Furthermore, let us say that a Beer Drinker drinks in the form of beer most of the alcohol he or she consumed during the past year. To be specific again, we shall say that at least 75 percent of this person's annual alcohol consumption must be beer. But of course, at this point, one could ask, what is beer? And we might decide beer is a malt beverage, with between 2 and 8 percent alcohol. We could continue to define malt beverage, but at some time we must fall back on our background of understanding.

The question is now whether the event has been clearly enough specified so that we could tell whether the next person entering the room was a Beer Drinker. The clarity test has been met when we are sure that this determination is no longer a matter of judgment, but a matter of
Notice that much of the activity of lawyers in writing a contract is to assure that there is clarity of

definition in each of the events of the contract. It will do no good to write in a contract that a product must be "of high quality" unless "high quality" is sufficiently defined and a procedure for measuring it adequately specified so that the event "high quality" can be determined in practice. To achieve clarity in event definition is a demanding process. However, it is necessary if we are to know what we are talking about as contrasted to what we are saying about it.

1.1.3 Usefulness

The question of whether the event is usefully defined as well as clearly defined raises an entirely different issue. There are many definitions that are clear but only some may be useful. In decision-making we pursue both event clarity and event usefulness to attain insight into the decision. For the moment, however, our goal is to focus on clarity.

1.1.4 The Possibility Tree

With this definition of Beer Drinker, we can depict the distinction of Beer Drinker by a diagram like Figure 1-1, called a possibility tree.

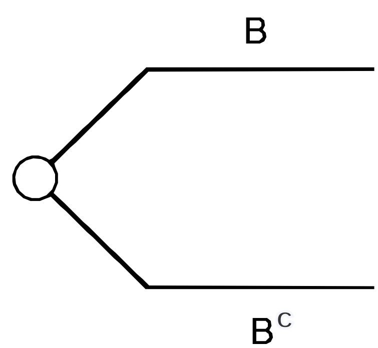

Figure 1-1. The Possibilities Represented by the Distinction Beer Drinker, B

Here we show a small tree with one branch representing the event "Beer Drinker" and denoted by B and another branch representing the event "not a Beer Drinker" and represented by B^c or "B complement". This diagram shows that we are considering two possibilities for the next person entering the room. That he or she will be a Beer Drinker and that he or she will not be a Beer Drinker.

Of course, being a Beer Drinker is not the only distinction we might make about the next person entering the room. Another might be whether he or she is a college graduate, an event we denote by G. After some thought we might agree that a clear definition of the distinction College Graduate is that the person be on the list of graduates of institutions that are accredited institutions providing two or more years of college education. If we further specified the accreditation agencies, then we would be even more sure that we had achieved clarity. Figure 1-2 shows the corresponding tree.

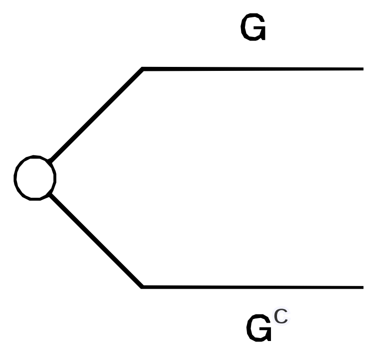

Figure 1-2. The Possibilities Represented by the Distinction College Graduate, G

The next person entering the room will either be a College Graduate or will not be by this definition.

1.1.5 Several Events or Kinds of Distinctions

Most decisions require possibilities composed of several events or kinds of distinction. For example, you might be interested in whether the next person entering the room is both a Beer Drinker and a College Graduate or a Beer Drinker and not a College Graduate etc. A tree diagram representing these multiple kinds of distinctions is shown in Figure 1-3.

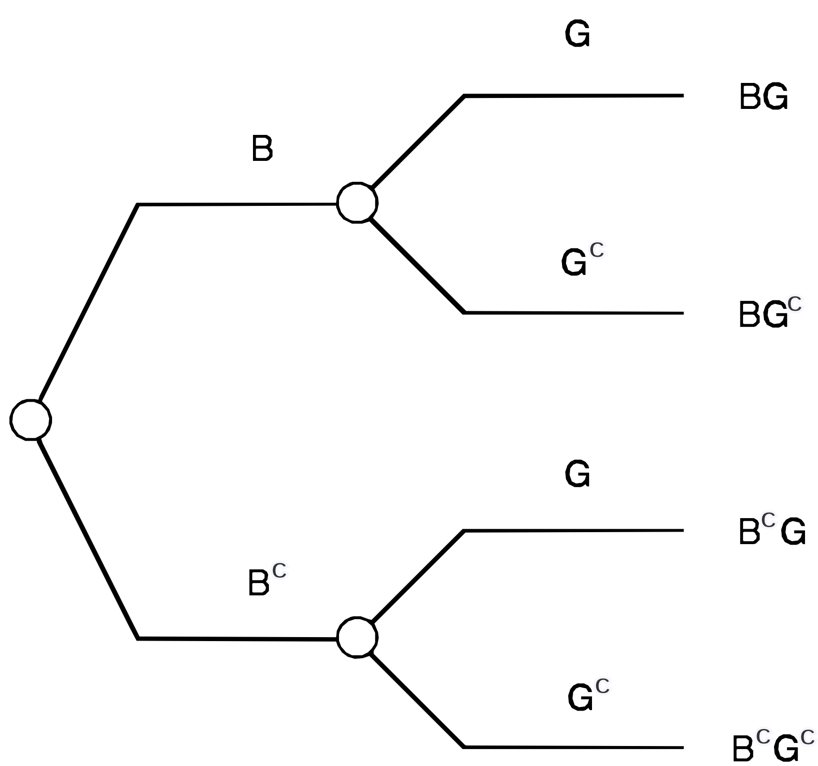

Figure 1-3. The Possibilities Represented by the Distinctions Beer Drinker, B, and College Graduate, G

The order of this tree indicates that we are first going to think about whether the person is a Beer Drinker and then whether the person is a College Graduate.

1.1.6 Elemental Possibilities

Each path through the tree represents one elemental possibility; this tree has four. We label the endpoint that characterizes each elemental possibility by writing together the sequence of letters that correspond to the events in the path. Thus the elemental possibility that the person is both a Beer Drinker and a College Graduate is represented by the top endpoint BG. The possibility that the person is neither corresponds to the bottom endpoint B^cG^c.

1.1.7 Mutual Exclusion

We observe that if one of the four elemental possibilities occurs then none of the other three can occur. By the design of the tree, no matter which two elemental possibilities we look at, if one occurs the other cannot occur. When two possibilities have this property we say that they are mutually exclusive; all elemental possibilities are mutually exclusive.

1.1.8 Collective Exhaustion

We also note that one of the elemental possibilities must occur. We describe this property by saying that the collection of elemental possibilities is collectively exhaustive. In brief, only one elemental possibility can occur and one must occur.

1.1.9 Compound Possibilities

The elemental possibilities can be used to construct compound possibilities. A compound possibility is a distinction represented by a collection of elemental possibilities. For example, the compound possibility represented by the distinction "either a Beer Drinker or a College Graduate, possibly both" would be represented by the top three endpoints in Figure 1-3. We describe such a compound possibility by joining with plus signs the symbols for the elemental possibilities it contains. Thus, this compound possibility is denoted by $BG \cup BG^c \cup B^cG$.

1.1.10 The Other "Or"

We can contrast this compound possibility with the one described by the distinction "either a Beer Drinker or a College Graduate, but not both." The first compound possibility we discussed is called the "inclusive or" because it includes the possibility of both. This one is the "exclusive or" because the possibility of both is excluded. Thus the notation for this compound possibility is simply $BG^c \cup B^cG$, the two elemental possibilities in the center of the tree. The possibility tree allows us to identify and eliminate the ambiguities that are created by everyday English phrases such as "either a Beer Drinker or a College Graduate."

1.1.11 Reversing the Order of Distinctions

Figure 1-3 shows the possibility tree where we have first thought about the distinction Beer Drinker and then the distinction College Graduate. There is no reason why we could not think first about whether the person was a College Graduate and then whether he or she was a Beer Drinker. If we draw the possibility tree in this order, we obtain Figure 1-4.

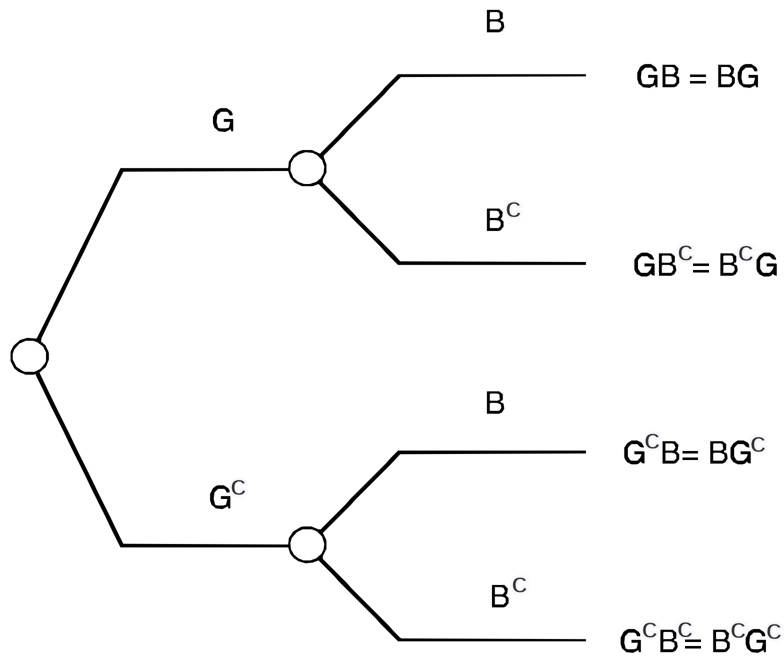

Figure 1-4. The Possibilities Represented by the Distinction College Graduate, G, Followed by the Distinction Beer Drinker, B

Note that this tree has the same elemental possibilities as that of Figure 1-3, but that the middle two elemental probabilities are switched in order in Figure 1-4. We also note that the order of letters in the notation for each endpoint is immaterial. The event of both a College Graduate and a Beer Drinker is the same as the event of both a Beer Drinker and a College Graduate.

If we draw Figure 1-4 from right to left, then we can put it in the form of Figure 1-5 which shows clearly that the same endpoints are produced by both trees.

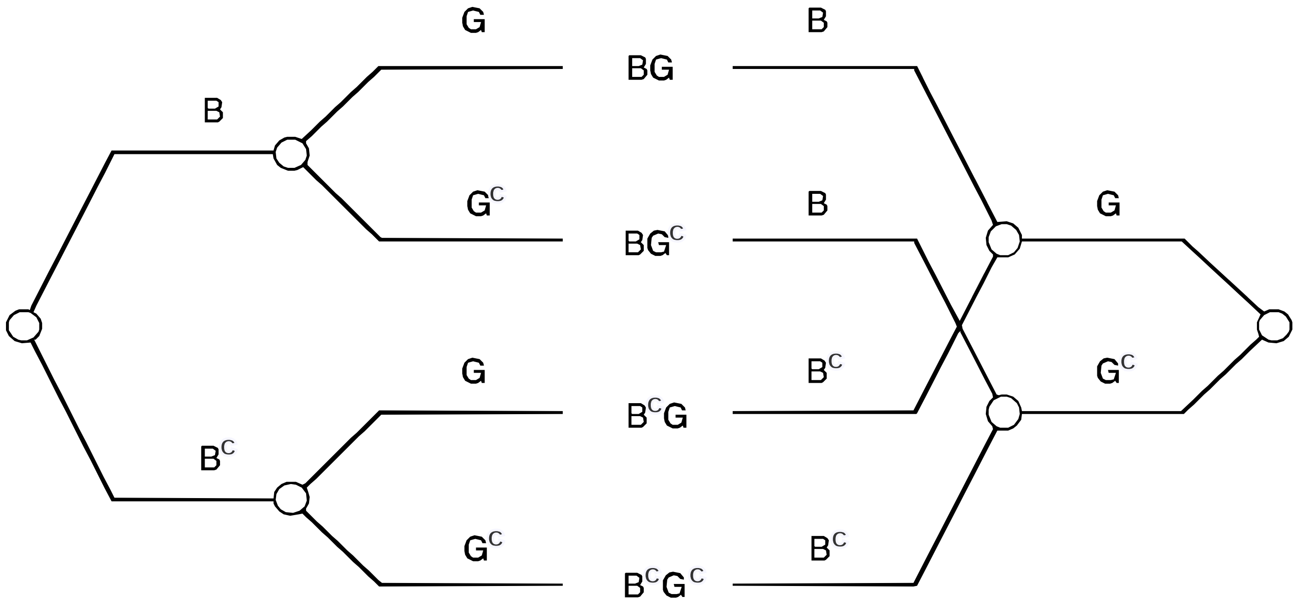

Figure 1-5. Construction to Show that the Same Possibilities Exist Regardless of Tree Order

1.1.12 Many Kinds of Distinction

We can introduce as many kinds of distinction as we like. For example, in a discussion about whether the next person entering the room drinks beer or has a college education, we might also want to introduce the sex of the person using the distinction Male or Female. Happily this distinction does not require an extensive clarity test discussion. We shall use M to designate Male and F to designate Female.

The possibility tree for these three distinctions appears as Figure 1-6.

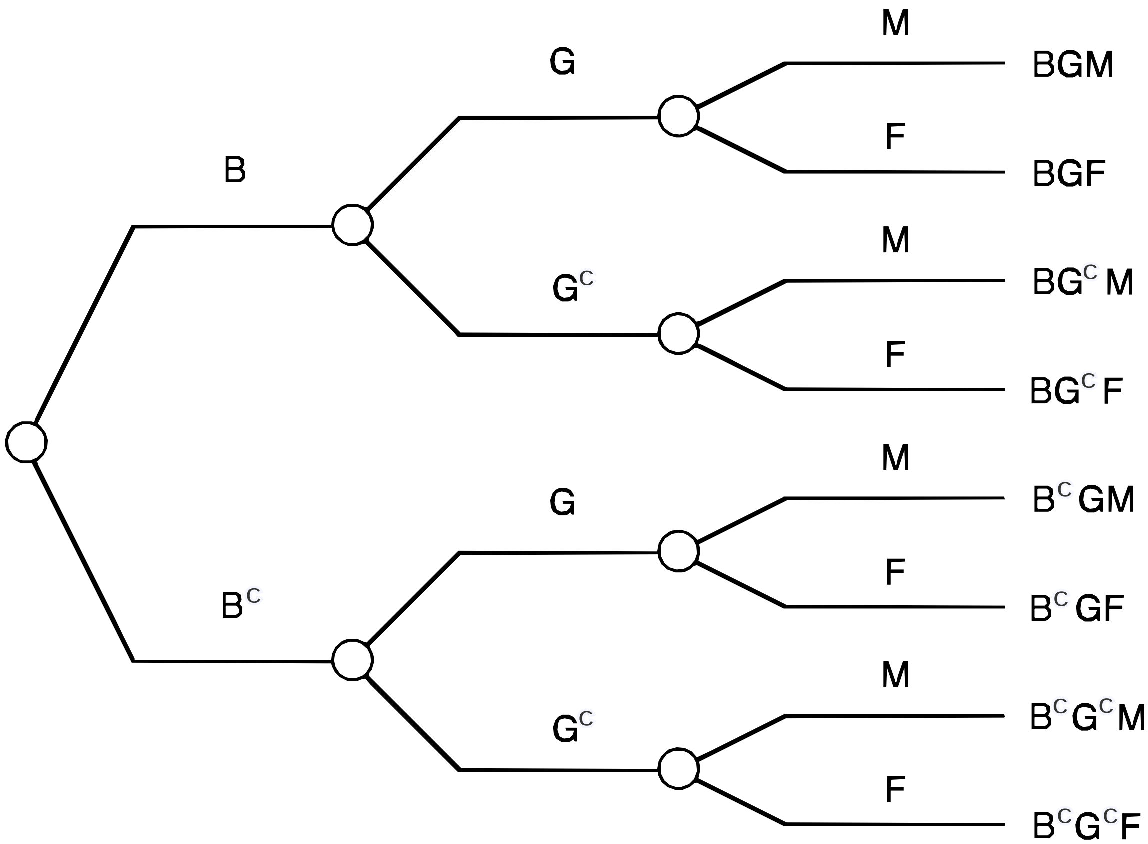

Figure 1-6. Result of Adding the Distinction of Sex

There are now eight elemental possibilities represented by the tree: BGM, BGF, BG^cM, etc. For example, the elemental possibility B^cG^cF represents the situation where the next person entering the room is a female who is not a Beer Drinker and not a College Graduate. In this possibility tree the compound possibility Male Beer Drinker could be represented by a collection of two elemental possibilities and denoted by BGM+BG'M. Thus, adding a kind of distinction does not prevent us from representing any compound possibility that could have been represented without it.

With three kinds of distinctions there are six possible orders for drawing the possibility tree, for any of the three kinds of distinction could be placed first and then any of the two others placed second. All orders are equally valid and produce the same endpoint elemental possibilities. If there are N kinds of distinctions, then there will be $N \cdot (N-1) \cdot (N-2) \cdots$ or $N!$ different orders for drawing the possibility tree.

1.1.13 Several Degrees of Distinction

We need not limit ourselves to creating simple dichotomies for each kind of distinction. We can create as many degrees of distinction, within a given kind, as we wish.

1.1.13.1 Degrees of Distinction for Beer Drinking

Suppose you believe that higher education is associated with less beer drinking, and that your companion disagrees. The disagreement has resulted in a bet about the next person entering the room. In defining the bet you may wish to deal with more than simply whether the next person entering the room is a Beer Drinker. You may want to distinguish three different levels of beer consumption, which we might denote by B_1, B_2, and B_3. Define B_1 as the event that the person is a light beer drinker, drinking no more than 50 quarts of beer per year; B_2, the event that the person is a medium beer drinker who drinks more than 50 quarts of beer per year but not more than 150 quarts per year; and B_3, the event that the person is a heavy beer drinker who consumes more than 150 quarts of beer per year. Thus every person who might enter the room can be characterized by one of the three events, B_1, B_2, or B_3. Figure 1-7 represents a possibility tree showing these three degrees of distinction for drinking beer.

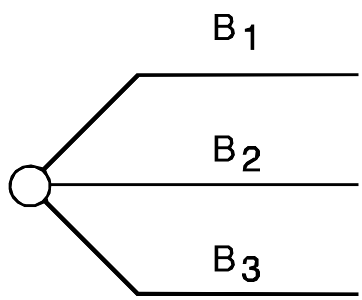

Figure 1-7. Three Degrees of Distinction for Drinking Beer

1.1.13.2 Degrees of Distinction for Education

In constructing our bet about beer drinking and education, we might also wish to create more degrees of distinction than simply whether the person was a College Graduate. Let us create four degrees of distinction for education, G_1 , G_2 , G_3 , and G_4. G_1 is the event that the person has completed no more than grade school; G_2 , the event that the person has completed more than grade school but no more than high school; G_3 , the event that the person has completed more than high school, but no more than undergraduate college; and G_4, the event that the person has completed more than undergraduate college. Figure 1-8 shows the corresponding possibility tree for the kind of distinction education and with the four degrees or levels of distinction that we have specified.

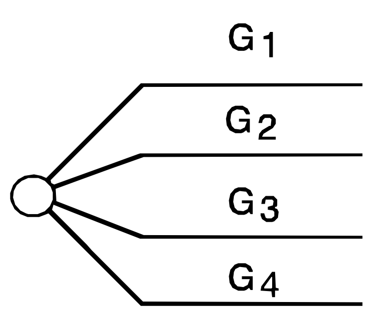

Figure 1-8. Four Degrees of Distinction for Level of Education

In creating the different degrees of distinction for both beer drinking and education we must assure that the clarity test has been passed.

1.1.14 Possibility Trees

In constructing the bet we will want to consider all the elemental possibilities generated by the two kinds of distinctions we have defined. Figure 1-9 shows the possibility tree that results.

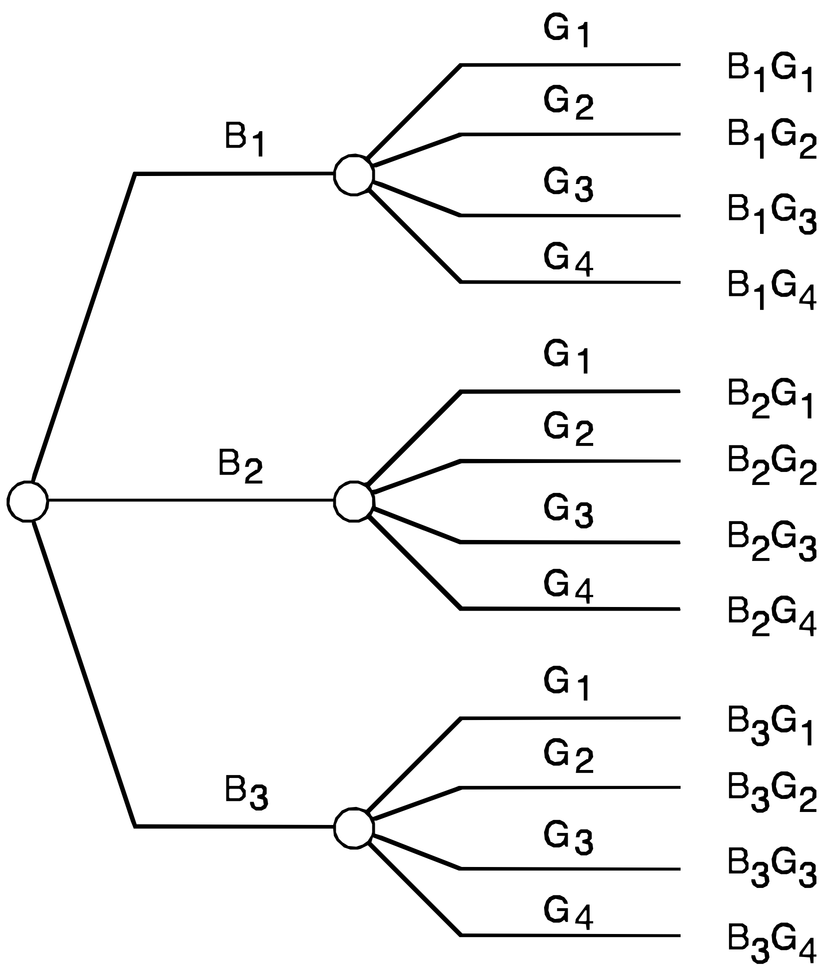

Figure 1-9. Possibility Tree with Multiple Degrees of Distinction for Two Kinds of Distinction

This tree shows the twelve elemental possibilities that we can distinguish, ranging from B_1G_1, the light beer drinker who has completed no more than grade school, to B_3G_4, the heavy beer drinker who has completed more than undergraduate college. In general, the number of elemental possibilities in a possibility tree is the product of the number of degrees of distinction of each kind; in this case, 3x4.

1.1.14.1 Distinction Order is Arbitrary

As with possibility trees composed of simple distinctions, possibility trees with multiple degrees of distinction can be drawn in any order; they will all generate the same endpoints. The endpoints will be mutually exclusive and collectively exhaustive regardless of tree order. Even when the number of kinds of distinction and degrees of distinction is so large that the tree is too large to be drawn, the concept of the tree of possibilities is still useful as an aid to visualization of the possibilities that have been created.

1.1.15 Simple Distinctions Are Sufficient

The introduction of several degrees of distinction of one kind is a useful, but not necessary generalization of the notion of simple distinctions or dichotomies that we have discussed. Several degrees of distinction of one kind can always be represented by several simple distinctions. For example, in our discussion of the amount of beer drinking, we could define two simple distinctions. The first distinction L represents a light beer drinker, a person who drinks no more than 50 quarts of beer per year. The second distinction H represents a heavy beer drinker, a person who drinks more than 150 quarts of beer per year. Figure 1-10 shows that the possibility tree generated by these two simple distinctions creates four elemental possibilities, LH, LH^c, L^cH, and L^cH^c.

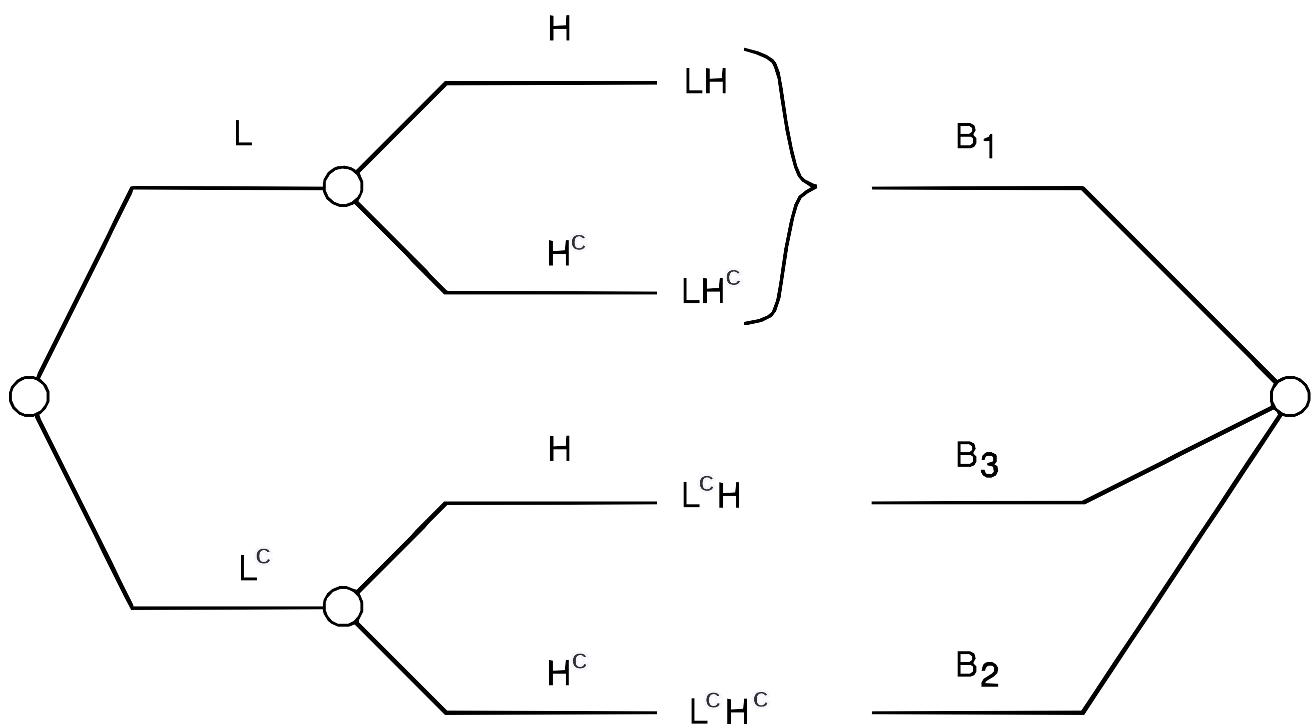

Figure 1-10. Multiple Degrees of Distinction Represented by Simple Distinctions

The figure shows that the elemental possibility L^cH^c is the degree of distinction B_2. The elemental possibility L^cH is the degree of distinction B_3. Finally, the elemental possibility LH and LH^c or $LH \cup LH^c$ represents the degree of distinction B_1. Here we note that LH is an elemental possibility that cannot occur because L and H as defined are mutually exclusive events. Therefore $B_1 = LH^c$, $B_2 = L^cH^c$, and $B_3 = L^cH$. The three degrees of distinction have been interpreted in terms of simple dichotomies. This procedure of introducing dichotomies be used repeatedly to allow any kind of distinction with multiple degrees to be represented simple distinctions.

1.1.16 Measures

In some settings like the bet we are discussing, it is useful to connect a number with each elemental possibility. We call this number a measure. This number in the present example could be your winnings on a bet based on the relation of education to beer drinking. Since you believe that people who drink beer heavily are less likely to be well-educated and you are betting with someone having an opposing view, you will win in situations where the next person entering the room drinks more beer and is less educated, or drinks less beer and is more educated. However, for middle levels of beer drinking and education, no money will change hands.

The specific bet is this: You will win \$300 if the next person entering the room is a heavy beer drinker with no more than a grade school education, B_3G_1. You will win \$200 if the person is a heavy beer drinker who has completed no more than high school, B_3G_2. You will win \$100 if the person is a light beer drinker who has completed more than undergraduate college, B_1G_4. However, you can also lose the bet. You will lose \$300 if the person is a heavy beer drinker who has completed more than undergraduate college, B_3G_4. You will lose \$200 if the heavy beer drinker has an undergraduate college education only, B_3G_3. Finally, you will lose \$100 if the person is a light beer drinker who has completed no more than grade school, B_1G_1. If the person entering the room is not in any of these categories there will be no payoff to either party to the bet.

The possibility tree with a measure representing your payoff from the bet is shown in Figure 1-11.

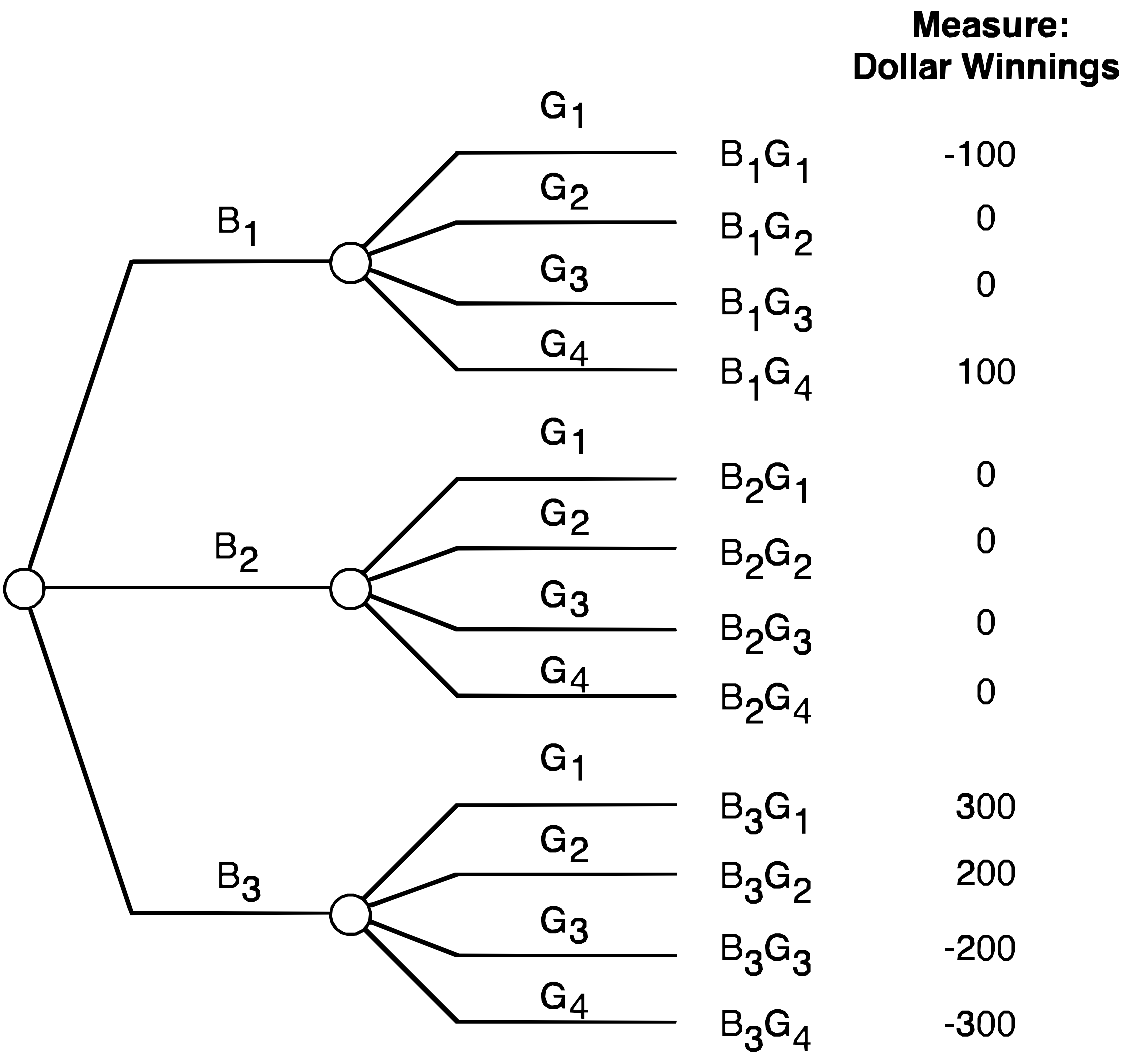

Figure 1-11. Possibility Tree with a Measure

We see that six of the elemental possibilities will yield no exchange of money, and that the others will have the payoffs we discussed.

1.1.16.1 Several Measures

Several different measures could be attached to each of the elemental possibilities. For example, your bet about beer drinking and education could involve not just money, but washing each other's cars some number of times. In this case we would attach number of car washes won or lost as a second column of measures at the end of the possibility tree.

The possibility tree to describe this situation is adequate when the next person entering the room can be characterized by an elemental possibility and the measure representing the exchange of money required by the bet for each elemental possibility is specified. Would it not be wonderful if all contracts were so clearly represented.

1.2 Probabilities

We have developed a framework for describing the possibilities that we have decided to include in our conversation. In other words, we have specified precisely what we are talking about. Now it is time to specify what we are saying about what we are talking about.

1.2.1 Describing Degree of Belief by Probability

Let us begin with the simple distinction Beer Drinker that we have previously defined and whose possibility tree appears in Figure 1-1. Now that we know what we mean by Beer Drinker, how sure are we that the next person entering the room will be a beer drinker?

1.2.1.1 Using All Information about the Specific Situation

The answer to this question depends on our general personal experience—everything we have learned throughout life up to this time. An important part of this experience is the setting to which we are applying our knowledge. The question is not what fraction of people are Beer Drinkers, but rather how likely it is that the next person entering this particular public room in this building, on this day, after this time will be a Beer Drinker. For example, in coming to the meeting room you might have noticed a raucous party in which a great deal of beer was being consumed. Or, you might have noticed that the building was host to a religious convention with many serious participants. You might also know that you have made an arrangement to meet a friend in this room in a few minutes and that this friend is a real beer lover. All information of this type is included in your total experience, your current state of knowledge. In a later section we will discuss the use of "knowledge maps" to make explicit how the individual elements of our state of knowledge might bear on the possibilities at hand.

1.2.1.2 Dividing Certainty

You can describe the likelihood that the next person entering the room will be a beer drinker by dividing your certainty between the two possibilities Beer Drinker, B and not a Beer Drinker, B^c. If, all things considered, you think it is four times as likely that the person will not be a Beer Drinker as be one, then you must divide your certainty in the ratio 1 to 4 between the two possibilities. Since you only have one unit of certainty, this means that you must place a fraction 0.2 on the possibility Beer Drinker and 0.8 on the possibility not a Beer Drinker. Figure 1-12 shows this division.

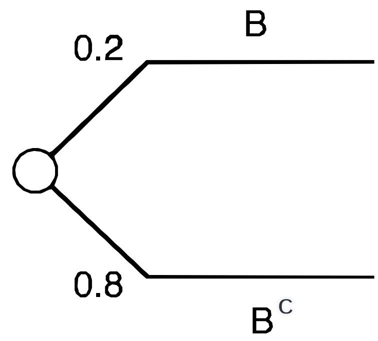

Figure 1-12. Division of Certainty on the Distinction Beer Drinker, B

The number 0.2 has been written adjacent to the upper branch and the number 0.8 adjacent to the lower branch. We call these two numbers probabilities. We say that the probability that the next person entering the room will be a beer drinker given the state of information represented by your total experience is 0.2.

1.2.1.2.1 Notation

Our notation for this statement is P$(B\,|\,\mathcal{E}) = 0.2$. The braces mean that you are making a probability assignment. The possibility on which you are making this assignment appears to the left of the vertical bar; in this case it is the event Beer Drinker, with symbol B. The state of knowledge on which the probability assignment is made appears to the right of the vertical bar. In this case it is simply your total experience which we denote by &. Your probability that the next person entering the room will not be a beer drinker is then P$(B^c\,|\,\mathcal{E}) = 0.8$.

You might also wish to describe what you know about whether the next person entering the room will be a College Graduate by dividing your certainty between the two possibilities G and G^c as shown in Figure 1-13.

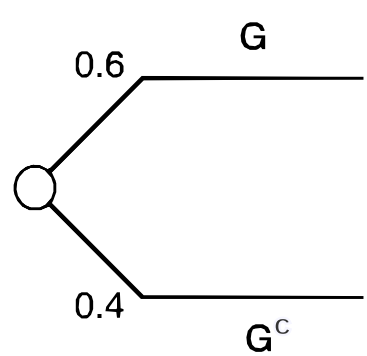

Figure 1-13. Division of Certainty on the Distinction College Graduate, G

Here you have assigned a probability 0.6 to the event College Graduate; P$(G\,|\,\mathcal{E}) = 0.6$; therefore, you have assigned P$(G^c\,|\,\mathcal{E}) = 0.4$. Thus you have divided your certainty between the two possibilities G and G^c in the ratio of 3 to 2.

1.2.1.3 Dividing Certainty Among Several Degrees of Distinction

The main rule of the probability game is that there is one unit of certainty to be divided among all the degrees of distinction. If we had several degrees of distinction for drinking beer, such as the three degrees of Figure 1-7, then the numbers on all branches must add to 1. Figure 1-14 shows the division of certainty you might have among the three distinctions B_1, B_2, B_3.

![Three Degrees(Figure1-14_bis.svg)
Figure 1-14. Division of Certainty Over Three Degrees of Distinction for Drinking Beer

A probability 0.7 is assigned to B_1, 0.2 to B_2, and 0.1 to B_3. This assignment says that since P$(B_1 \mid \mathcal{E})$ = 0.7 and P$(B_3 \mid \mathcal{E})$ = 0.1, then the next person entering the room is, in your opinion, seven times as likely to be someone who drinks less than 50 quarts of beer per year as to be a person who drinks more than 150 quarts of beer per year.

If you wished to use the more refined levels of education we have previously defined, then you might divide your certainty as shown in Figure 1-15.

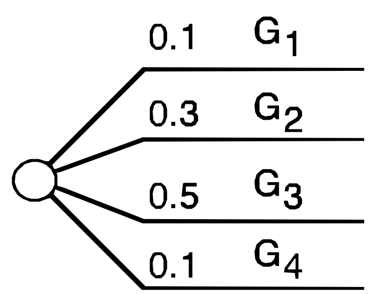

Figure 1-15. Division of Certainty Over Four Degrees of Distinction for Level of Education

Since the event $G = G_3 \cup G_4$, the two branches corresponding to G_3 and G_4 in the figure must receive a total of 60 percent of your certainty. The probability assignment 0.1, 0.3, 0.5, 0.1, to G_1, G_2, G_3, G_4 is consistent with your beliefs as expressed in Figure 1-13. Since P$(G_3 \mid \mathcal{E}) = 0.5$, you see it as likely as not that the person has an undergraduate degree, but no higher degree.

1.2.2 The Probability Tree

The purpose of the Beer Drinker and College Graduate distinctions was to discuss the connection between being a beer drinker and being a college graduate. To represent your beliefs about these distinctions, you attach probabilities to the possibility tree of Figure 1-3 to obtain the probability tree of Figure 1-16.

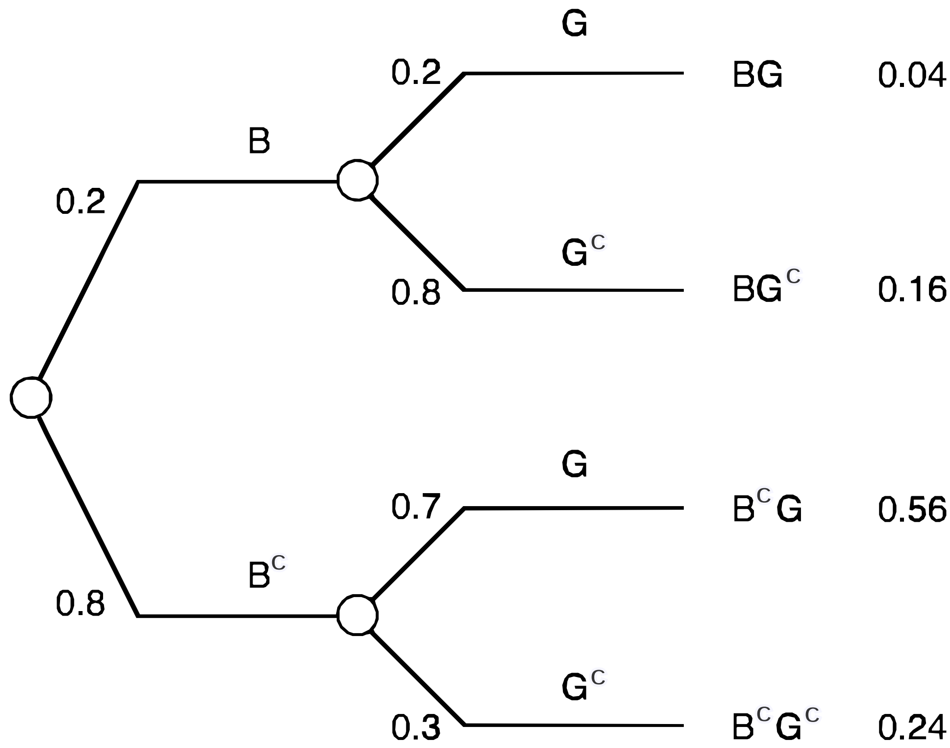

Figure 1-16. Probability Tree: Division of Certainty by Two Successive Distinctions

The beginning of the tree shows your division of certainty according to the probabilities of 0.2 and 0.8 on the events B and B^c, as in Figure 1-12. But now we must consider further division of your certainty. The B branch of the tree represents the possibility that you know that the person is a Beer Drinker. If you knew this were true, how would you then split your certainty between the events College Graduate and not a College Graduate, G and G^c? Figure 1-16 indicates that in this case you would split your certainty in the ratio 1 to 4 and hence assign a probability 0.2 to the person's being a College Graduate and 0.8 to the person's not being a College Graduate. Similarly, the split following the branch B^c shows that you would assign a probability of 0.7 to the person's being a College Graduate and 0.3 to the person's not being a College Graduate if you knew that the person was not a Beer Drinker.

1.2.2.1 Conditional Probabilities

We call the probabilities that we assigned to G and G^c given that we know that either B or B^c occurred "conditional probabilities", for they are conditional on the occurrence of the first distinction. Any conditional probability is conditioned on all events that preceded it in the possibility tree. The notation for conditional probabilities follows the same rule for notation that we defined previously; the conditioning event is included to the right of the vertical bar. Thus, the probability that the person will be a College Graduate given that the person is a Beer Drinker is written as P$(G \mid B, \mathcal{E})$, and have the value 0.2. The probability of College Graduate given not a Beer Drinker is written P$(G \mid B^c, \mathcal{E})$, and has the value 0.7.

1.2.2.2 Elemental Probabilities

Figure 1-16 now shows that your certainty has been split twice. Once by the distinction Beer Drinker and once by the distinction College Graduate. If 20 percent of your certainty flows through the branch B and 20 percent of that flows through the branch G, then the fraction of the original certainty represented by the elemental possibility BG is 0.04. This is the probability that the person is both a Beer Drinker and a College Graduate, written P$(BG \mid \mathcal{E})$. For brevity, we call the probability of an elemental possibility an elemental probability. The four elemental probabilities are shown at the endpoints of the tree. They are in each case obtained by multiplying the branch probabilities on all branches leading to the endpoint. Since we started with one unit of certainty, the sum of all elemental probabilities must be 1.

Since P$(B^cG \mid \mathcal{E}) = 0.56$ is the largest of the four elemental probabilities, we see that the most probable elemental possibility in this case is that the next person entering the room will not be a Beer Drinker but will be a College Graduate. Furthermore, since the probability of B^cG is larger than 0.5, B^cG is more likely than the three other elemental possibilities combined. Note that the event College Graduate is composed of the elemental possibilities BG and B^cG. When P$(B^cG \mid \mathcal{E})$ is added to the elemental probability of BG, which is 0.04, the total is 0.6. This is precisely the probability you assigned to the event that the person be a College Graduate as recorded in Figure 1-13. Thus, you have been consistent in your probability assignments.

1.2.3 Probability Tree Reversal

The probability tree you have created in Figure 1-16 shows that you think there is a strong connection between being a Beer Drinker and being a College Graduate. You have said that if a person is a Beer Drinker, that person has an 80 percent chance of not being a College Graduate, but that if the person is not a Beer Drinker, then the person has a 70 percent chance of being a College Graduate. But suppose you knew that the person was a College Graduate, how likely is the person to be a Beer Drinker?

1.2.3.1 Reversing the Possibility Tree

We can answer this question by reversing the order of the distinctions as we did in the possibility tree of Figure 1-4. Remember that reversing the order of distinctions creates the same elemental possibilities. The reversed possibility tree has the order College Graduate – Beer Drinker and is shown in Figure 1-17.

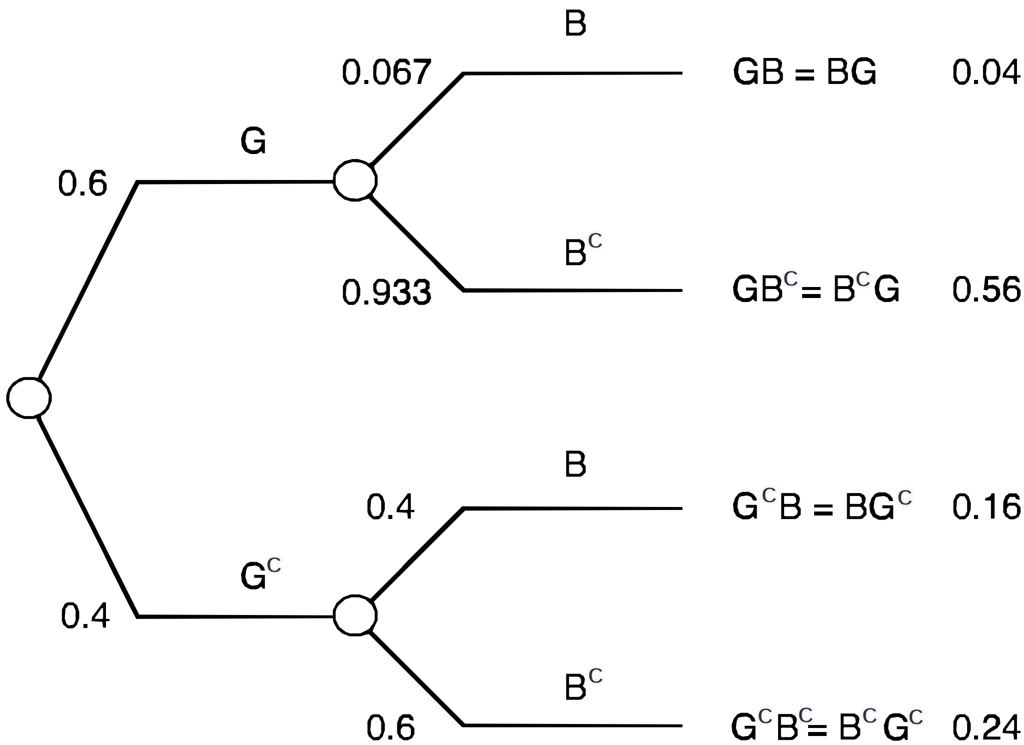

Figure 1-17. Reversing the Order of Distinctions in a Probability Tree

1.2.3.1.1 Copying the Elemental Probabilities

Since Figure 1-17 has the same elemental possibilities as the probability tree in Figure 1-16, and since reversing the order should not by itself change the division of certainty among the elemental possibilities, we can can copy the elemental probabilities associated with the elemental possibilities of Figure 1-16 to the corresponding position for those possibilities in Figure 1-17. As we demonstrated in Figure 1-5, this will require reversing the two middle elemental possibilities.

1.2.3.1.2 Reconstructing the Divisions of Certainty

Once we know how much of your certainty has arrived at each endpoint of the tree, it is a simple matter to construct all the probabilities in the tree. For example, since whatever certainty flowed through branch G must end up at either the endpoint GB or GB^c the amount of certainty that went down that branch must be the sum of 0.04 and 0.56 or 0.6. We write this number as the probability of G associated with the initial division of certainty. Recall that we previously commented on your consistency in having assigned a 60 percent chance to the person's being a college graduate. Similarly, the 0.4 on the branch G^c is the sum of the probabilities 0.16 and 0.24 associated with the elemental possibilities G^cB and G^cB^c. We now proceed to obtain the conditional probabilities B and B^c given G and G^c.

1.2.3.1.3 Finding the Conditional Probabilities

Let us begin with the conditional probabilities following the branch G. Here we know that your certainty was split in such a way that the upper endpoint received a probability 0.04 and the lower one 0.56. This means that it must have been split in the ratio 1-14. Since the sum of the conditional probabilities must be 1, this means you must assign a conditional probability 1/15 or 0.067 to the person's being a Beer Drinker given that the person is a College Graduate. In our notation, we write P$(B \mid G, \mathcal{E}) = 1/15 = 0.067$. Of course, this means that the conditional probability that the person will not be a Beer Drinker given that the person is a College Graduate, P$(B^c \mid G, \mathcal{E})$, is $14/15 = 0.933$.

Continuing the conditional probability assignment to the branches that follow G^c, we see that the conditional probabilities on the two branches must be in the ratio 0.16 to 0.24, which means that they must be 0.4 and 0.6. Thus, the probability that the person will not be a Beer Drinker if the person is not a College Graduate, P$(B^c \mid G^c, \mathcal{E})$, is 0.6.

1.2.3.2 Interpreting the Results

This reversed probability tree shows the probabilities that you must assign to the distinction Beer Drinker if you know the results of the distinction College Graduate. For example, we see that if you know that the person is a College Graduate, you assign a 14/15 chance that the person will not be a Beer Drinker. And if you know that the person is not a College Graduate, then the person has a 60 percent chance of not being a Beer Drinker. Figure 1-17 contains no information that was not present in Figure 1-16. The information has merely been processed in a new way to show more completely the connection you make between the two distinctions. We speak of this connection as the relevance of one distinction to another.

1.2.4 Relevance

We say that the simple distinction E–E^c is relevant to the simple distinction F–F^c if the probability of F given that E occurs is not equal to the probability of F given that E does not occur. Or in our notation P$(F \mid E, \mathcal{E})$ ≠ P$(F \mid E^c, \mathcal{E})$. Relevance is thus defined by the requirement that given your information and knowing of the occurrence of one distinction will change the probabilities you assign to the other distinction. On reflection you will see that this definition corresponds to our commonsense understanding of relevance. We say that education is relevant to income because we assign higher probabilities to higher incomes if we know the person is well educated than if we don't. We say that flipping a coin is not relevant to the weather because we would assign the same probabilities to different weather states regardless of whether Heads or Tails appeared.

1.2.4.1 Causality

We must emphasize that relevance does not imply any claim of causality. Consider the distinctions that it is raining outside and that passersby are carrying umbrellas. Most people would think that the rain caused the passersby to carry umbrellas and not vice versa, but this thought has no place in our discussion. If you assign higher probabilities to the event it is raining outside when you are told that passersby are carrying umbrellas, then passersby carrying umbrellas is relevant to whether it is raining.

1.2.4.2 Relevance Depends on Knowledge

We cannot emphasize too strongly that relevance rests on your state of information at a particular time and not on the definition of the events. People who are talking about the same two events may differ on whether one is relevant to the other. Relevance is not a matter of logic, but of information. In my experience most errors in dealing with uncertainty are the result of misunderstanding or misapplying the concept of relevance.

1.2.4.3 Mutual Relevance

If event E is relevant to event F, is event F relevant to event E? Sometimes? Always? Never? After reflecting on the question, take a look at Figure 1-18.

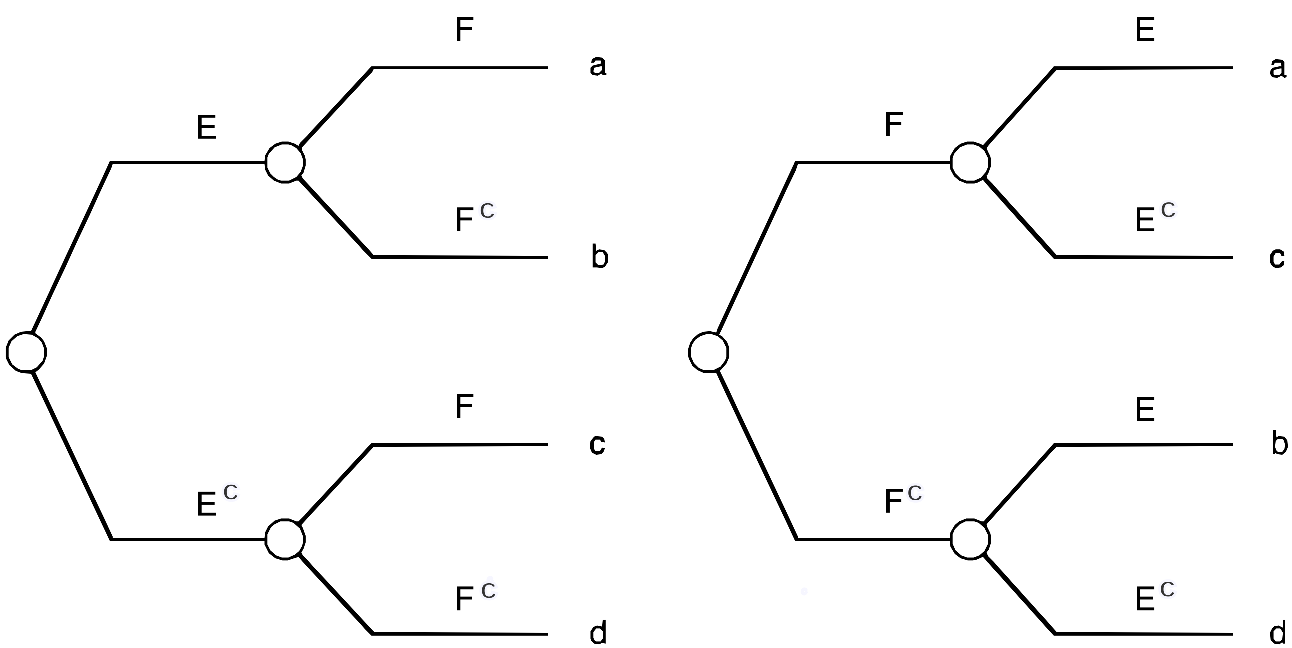

Figure 1-18. Proof that if One Distinction is Relevant to a Second, the Second is Relevant to the First

On the left of the figure, we have shown a probability tree for the distinctions E and F. We have not shown the branch probabilities but instead simply written the elemental probabilities at the endpoints of the tree; they are given by a, b, c, and d as shown. If, as hypothesized, the simple distinction E is relevant to the distinction F, then the probability of F given that E occurs must be different from the probability of F that given that E does not occur. This means that the ratio of a to b is not equal to the ratio of c to d or, equivalently, ad ≠ bc. We can show this in detail as

```
a/(a+b) ≠ c/(c+d)

a(c+d) ≠ c(a+b)

ac + ad ≠ ac + bc

ad ≠ bc

a/b ≠ c/d
```

Now examine the reversed probability tree on the right side of the figure. The only difference is that the elemental probabilities assessed on the two middle elemental possibilities have been reversed to be placed in their proper position. If F is to be relevant to E, then the probability of E given F must not be equal to the probability of E given F',

```
a/(a+c) ≠ b/(b+d)
```

To show this, we recall ad ≠ bc, and add ab to both sides,

```
ab + ad ≠ ab + bc

a(b+d) ≠ b(a+c)

a/(a+c) ≠ b/(b+d)
```

as we wished to show.

Therefore, we have proved that if one simple distinction is relevant to another, then the second is relevant to the first. Relevance is a mutual property.

Another way of expressing relevance is to say that if E is relevant to F, then P$(F \mid E, \mathcal{E}) \neq$ P$(F \mid \mathcal{E})$. The probability you assign to F given that E occurs is different from the probability you would assign to F if you had no knowledge of E. In our tree this would require,

```
a/(a+b) ≠ (a+c)/(a+b+c+d)

aa + ab + ac + ad ≠ aa + ab + ac + bc
```

and, therefore,

```
ad ≠ bc
```

which we know is the case. The two definitions of relevance are equivalent for simple distinctions.

Observe that the only way you can determine whether distinction E is relevant to distinction F in a probability tree is by examining the numbers on the tree in detail to see whether the conditional probabilities on F are the same given either E or E^c. The probability tree is not the most convenient graphical representation of relevance.

1.2.4.4 The Relevance Diagram

To represent relevance we use a relevance diagram. A relevance diagram contains a circular or oval node for each distinction. If the probability tree is going to be drawn in the order E–F, then an arrow would be drawn from E to F. The left side of Figure 1-19 shows the corresponding relevance diagram.

Figure 1-19. A Pair of Relevance Diagrams

If we were to assert that E was not relevant to F, then we would omit the arrow and simply leave E and F as unconnected nodes.

1.2.4.4.1 Reversing Arrows

In view of our discussion of mutual relevance, if we have a relevance diagram in the form shown on the left of Figure 1-19, we can always convert it into the form shown on the right. This simply means that the first distinction in the probability tree will be F and the second, E. The probability calculations required to produce consistency in the two representations are exactly those we performed in constructing the probability tree of Figure 1-17. Notice that such tree or arrow reversal is possible because all the probability assignments in the tree were based on the same state of information, in this case, &. We shall find that the relevance diagram is an extremely useful tool for representing the assertions made in probability assignment. As we continue our discussions, we shall use such diagrams in larger form and with more features as powerful tools for representing problems of uncertainty and decision. However, for the Beer Drinker—College Graduate distinctions, the relevance diagram has either of the simple forms shown in Figure 1-20.

Figure 1-20. Relevance Diagrams for Beer Drinker/College Graduate

1.2.5 The Relevance of Marijuana Use to Heroin Use

The question of relevance can have important implications in everyday life. Some years ago a defendant convicted of selling marijuana appeared before a judge for sentencing. The judge said, "I have observed in my many years on the bench that most heroin addicts began by u[illegible] marijuana. Since the drug you sell leads on to heroin addiction, I am going to sentence you [illegible] you were a heroin dealer." The judge then gave the convict a long sentence.

1.2.5.1 Event Definitions

Let us examine the judge's reasoning in some detail. We will include as part of our state of information and that we are talking about American adults, those over 18 years of age. Let us define M as the event that a person is a Serious Marijuana User using the criterion that he or she must have consumed at least 100 marijuana joints during his or her lifetime. Let us define H as the event that the person is a Serious Heroin User, specified as having ingested at least 20 doses of heroin over a lifetime.

1.2.5.2 Probability Assignments

We then construct the probability tree in the order M–H as shown on the left side of Figure 1-21.

Figure 1-21. The Relevance of Marijuana Use to Heroin Use

The probability assignments shown are those typical of a group in which this situation is discussed. We see that there is a probability 0.1 that an American adult will be a Serious Marijuana User by our criterion, and if the person is a Serious Marijuana User then there is a probability of 1 in 100 of being a Serious Heroin User. However, if the person is not a Serious Marijuana User by our criterion, then the probability of being a Serious Heroin User is 1 in 1,000. The four elemental probabilities implied by the tree are shown at the ends of the branches as usual. We see for example, that the elemental possibility MH has a probability of 1 in 1,000.

1.2.5.2.1 Reversing the Tree

Now let us reverse the probability tree as shown on the right side of Figure 21. With elemental probabilities recorded in their appropriate position, we can compute the probability assignments in this tree that will make it consistent with the original one. The probability of the event H is the sum of the probabilities of the elemental possibilities HM and HM^c or .0019; the event H^c has probability .9981. The conditional probabilities are obtained from the ratio of the elemental probabilities and the requirement that they add to 1. Thus, the probability of M given H, P$(M \mid H, \mathcal{E}) = 0.0010/(0.0010 + 0.0009)$ or 0.526; the conditional probability P$(M^c \mid H, \mathcal{E}) = 0.474$. Similarly we find P$(M \mid H^c, \mathcal{E}) = 0.099$ and P$(M^c \mid H^c, \mathcal{E}) = 0.901$.

1.2.5.2.2 Interpretation

Then we can interpret what we have computed. We see first that the probability that an adult American will be a Serious Heroin User by our definition is 0.0019 or about two thousandths. Given that the person is a Serious Heroin User, the probability of being a Serious Marijuana User is 0.526. This is consistent with the judge's observation that when he interviewed heroin addicts, he found that many of them were or had been heavy users of marijuana. The probability of marijuana use given heroin use is sizeable. However, from this observation the judge had concluded that marijuana use leads to heroin use. This clearly is not the case in our example, since the probability $P(H \mid M, \mathcal{E})$ shows that only a 1 in 100 chance of a Serious Marijuana User becoming a Serious Heroin User. The fact that the probability of M given H is large does not imply that the probability of H given M is large. The judge incorrectly concluded that marijuana use is likely to lead on to heroin use from the correct observation that a majority of heroin users were or had been also marijuana users.

1.2.5.3 Distinctive versus Associative Logic

Improper reasoning by confusing the order of conditional probabilities is both pervasive and subtle. Without the foundation in logical thought that we are presenting, it is difficult for a person to see why he has made a logical error. It is as if the person used a simple associative, nondirectional logic to characterize the world rather that the precise, distinctive logic we have described. Thus, in many people's minds, the notion that heroin use "goes with" marijuana use allows them to act as if both of the conditional probabilities M given H and H given M are large, when we have seen that this need not be so.

1.2.6 Males and Hemophilia

To drive home the fact that a conditional probability and its inverse can be very different, let us consider two events M and H with different definitions. Let M be the event that the person is a male, and H be the event that the person is a hemophiliac. The probability of H given M, that a person will be a hemophiliac given that he is a male is very small, 1 in 1,000 or less. However, the probability of M given H is virtually 1, since only males appear to express the disease of hemophilia.

I once presented this discussion of both the marijuana/heroin and male/hemophilia example to a group of medical doctors who had returned to the university for advanced fellowship work. One of them commented that this was certainly an important logical observation, but that such errors in thought would not be common in people like themselves who had been thoroughly trained in diagnosis and treatment.

1.2.7 The Relevance of Smoking to Lung Cancer

At this point we began to discuss the question of the relation of smoking to lung cancer.

1.2.7.1 Formulation

We defined the event being a Heavy Smoker, S, as having smoked at least two packs of cigarettes per day for a period of at least 10 years during a lifetime. We defined the event contracting Lung Cancer, L, according to the usual medical definition, well understood by th
group. Then one of the doctors, who was not a specialist in the treatment of lung cancer,

assigned the probabilities shown in the probability tree at the left side of Figure 1-22, with the understanding that we were talking about adult Americans, as before.

Figure 1-22. The Relevance of Smoking to Lung Cancer

He assigned 1 chance in 4 that a person would be a Heavy Smoker, and if he or she were a Heavy Smoker S that the chance of getting lung cancer was about 0.1. If the person was not a Heavy Smoker S^c then the chance of getting lung cancer was 1 in 100. The elemental probabilities were obtained by multiplying the probabilities along each branch and recorded as shown. When transferred to the tree in reverse order, shown on the right side of Figure 1-22, they produced about a 3 percent chance of Lung Cancer and a 0.769 chance of being a Heavy Smoker given that a person did get Lung Cancer.

1.2.7.1.1 Interpretation

The group then discussed the assignments that the doctor had made. A lung cancer specialist remarked that he thought that the probability of L given S, assigned at 0.1, was much too low. When asked why, he responded "because when I visit my lung cancer ward, it is full of smokers." This observation, of course, merely reflects what the first doctor had said, because the probability of S given L is, in fact, 0.769, so he, too, would expect that a majority of patients in a lung cancer ward would be heavy smokers by our definition. In fact, the probability of getting Lung Cancer if you are a Heavy Smoker is nowhere near as high as the probability of being or having been a Heavy Smoker given that you have Lung Cancer.

When even highly-trained people fall victim to inferior reasoning, we know that we all must be careful. No one knows how many people are spending extra years in jail or perhaps being improperly treated in hospitals because of a failure to grasp the difference between distinctive and associative logic.

1.2.8 Probability Trees for Several Kinds of Distinction

We can construct probability trees for as many kinds of distinctions as we like. For example, if we add the sex of the person entering the room to our existing distinctions as we did in the possibility tree of Figure 1-6, we can construct a corresponding probability tree as in Figure 1-23.

Figure 1-23. Probability Tree for Three Distinctions

The probability assignments for the first two distinctions Beer Drinker and College Graduate are exactly those shown in Figure 1-16. All that you must add are the conditional probability assignments on the sex of the person entering the room conditional on the path through the distinctions Beer Drinker and College Graduate. The tree shows that you have assigned th probability that the person entering the room will be Male given that he is a both Beer Drink and a College Graduate as P$(M \mid BG, \mathcal{E}) = 0.8$, and therefore that P$(F \mid BG, \mathcal{E}) = 0.2$ and have also assigned $\mathsf{P}(M \mid BG^c, \mathcal{E}) = 0.7$, $\mathsf{P}(M \mid B^cG, \mathcal{E}) = 0.6$ and $\mathsf{P}(M \mid B^cG^c, \mathcal{E}) = 0.3$. These assignments reflect your belief that in this situation as contained in $\mathcal{E}$, being a Beer Drinker or being a College Graduate or both are positively relevant to the event of being a male.

1.2.8.1 Generating Elemental Probabilities

The eight elemental possibilities represented by the endpoints of this tree have probabilities obtained by multiplying the branch probabilities along the path that leads to each endpoint. For example, the elemental possibility BG^cM referring to a Beer Drinker –non-College Graduate– Male has probability $\mathsf{P}(BG^cM \mid \mathcal{E}) = 0.112 = \mathsf{P}(B \mid \mathcal{E})\mathsf{P}(G^c \mid B, \mathcal{E})\mathsf{P}(M \mid BG^c, \mathcal{E})$. Notice the pattern in generating this elemental probability. It is composed of the probability of the first distinction times the probability of the second distinction given the first times the probability of the third distinction given the first and second. We shall find it useful to place this result in general form.

1.2.8.2 The Chain Rule for Distinctions

Suppose that we have many distinctions A, B, C and several degrees of distinction any one of which we designate by A_i, B_j, and C_k. Then in accordance with our visualization of the probability tree, we can write for the probability of the elemental possibility $A_iB_jC_k$,

$$\mathsf{P}(A_iB_jC_k \mid \mathcal{E}) = \mathsf{P}(A_i \mid \mathcal{E})\mathsf{P}(B_j \mid A_i, \mathcal{E})\mathsf{P}(C_k \mid A_iB_j, \mathcal{E})$$

We call this the chain rule for distinctions. The chain rule shows how to compute the probability of any elemental possibility as the product of conditional probabilities. Since the tree that generates the elemental possibility $A_iB_jC_k$ could be drawn in any of the six possible orders we discussed, the chain rule for three events can be written in six possible ways. For example, this elemental probability is also given by,

$$\mathsf{P}(A_iB_jC_k \mid \mathcal{E}) = \mathsf{P}(C_k \mid \mathcal{E})\mathsf{P}(A_i \mid C_k, \mathcal{E})\mathsf{P}(B_j \mid A_iC_k, \mathcal{E})$$

We shall soon have more to say about probability tree reordering.

1.2.8.3 Probabilities of Compound Events

The probability of any compound event can be obtained by adding the probabilities of the elemental possibilities of which it is composed. Thus, if we wish to find the probability that the next person entering the room will be a Male using the probability tree in Figure 1-23, we note that the event Male is composed of four elemental possibilities,$ M = BGM \cup BGM \cup B^cGM \cup B^cG^cM$; therefore,

$$\mathsf{P}(M \mid \mathcal{E}) = \mathsf{P}(BGM \mid \mathcal{E}) + \mathsf{P}(BG^cM \mid \mathcal{E}) + \mathsf{P}(B^cGM \mid \mathcal{E}) + \mathsf{P}(B^cG^cM \mid \mathcal{E}) = 0.032 + 0.112 + 0.36 + 0.072$$

Thus implicit in your probability assignments of Figure 1-23 is the assignment of probability 0.552 to the event that the next person entering the room in this situation will be a Male.

Suppose that your real purpose all along had been to assign this probability. Suppose further that you found it easier to assign the probability conditional on whether the person was a Beer Drinker and conditional on whether the person was a College Graduate. Then you would have constructed precisely the probability tree of Figure 1-23 as an aid to making the probability assignment on Male. We would describe your actions as introducing the distinctions Beer Drinker and College Graduate into the discussion to aid your thought about the distinction Male–Female. Such expansion of the conversation is a fundamental tool of thought.

1.2.8.4 Expanding the Conversation

Suppose we have the distinction A discussed above, with typical level A_i, and that we wish to assign the probability P$(A_i \,|\, \mathcal{E})$. If we find it easier to think about A_i by introducing another distinction B with typical level \$ B_j
, $thenwecoulddrawaprobabilitytreeintheorderB - Aandobtaintheprobabilityof\$A_i$ by adding the probabilities of the elemental possibilities that constitute the compound event A_i in this tree,

$$\mathrm{P}(A_i \,|\, \mathcal{E}) = \Sigma_j \mathrm{P}(A_i B_j \,|\, \mathcal{E}) = \Sigma_j \mathrm{P}(B_j \,|\, \mathcal{E}) \mathrm{P}(A_i \,|\, B_j, \mathcal{E})$$

This result is equivalent to applying the chain rule to compute each of the elemental probabilities and then summing them over the possible levels of B.

If we wish to introduce the further distinction C with typical level C_k, then we would write,

$$\mathrm{P}(A_i \,|\, \mathcal{E}) = \Sigma_j \Sigma_k \mathrm{P}(A_i B_j C_k \,|\, \mathcal{E})) = \Sigma_j \Sigma_k \mathrm{P}(C_k \,|\, \mathcal{E}) \mathrm{P}(B_j \,|\, C_k, \mathcal{E}) \mathrm{P}(A_i \,|\, B_j C_k, \mathcal{E})$$

Here we have introduced two distinctions B and C into the discussion, computed the elemental probabilities using the chain rule, and then summed over all elemental possibilities contained in the compound possibility A_i to obtain the probability we desire. We use this general method of introducing new distinctions into the conversation in virtually every application of decision analysis.

1.2.9 Relevance Diagrams

We can represent the probability tree of Figure 1-23 by the relevance diagram in Figure 1-24.

Figure 1-24. A Relevance Diagram for Three Distinctions

The rule for such a diagram is that the arrows entering a distinction node show the distinctions upon which the probability assignment for that node is conditioned. If a node has no input arrows, then it is conditioned only on the general state of information &. Thus in Figure 1-24 we see that the Beer Drinker node has no input and therefore its probability assignment has no conditions. The College Graduate node is conditioned on the Beer Drinker node. The sex distinction node is conditioned on both the Beer Drinker and College Graduate distinction. This conditioning corresponds exactly to that used in assigning probabilities in the probability tree of Figure 1-23.

1.2.9.1 Alternate Assessment Orders

As we know, there are $N!$ orders for a possibility tree with N distinctions. The distinctions B, G, S can therefore be ordered in six ways; Figure 1-25 shows the six possible associated relevance diagrams.

Figure 1-25. Relevance Diagrams for Three Distinctions

The one in the upper left corresponds to the probability tree of Figure 1-23. Once we have assigned the probability for any one of these diagrams, we can deduce the probabilities for any other. Let us convert the diagram we have assigned to the relevance diagram that appears on the right of the second row in Figure 1-25. The assessment order for this diagram is first Sex, then Beer Drinker given Sex, and finally College Graduate given both Beer Drinker and Sex.

1.2.9.2 Constructing the Probability Tree for a Specified Assessment Order

We shall now show how to compute the probability tree corresponding to this relevance diagram, a tree shown in Figure 1-26.

Figure 1-26. Reordered Probability Tree for Three Distinctions

Our first step in creating the probability assignments is to record at the appropriate endpoints the elemental probabilities from the probability tree of Figure 1-23. Thus, the path MB^cG in Figure 1-26 has a probability 0.336 since it is the probability of the endpoint B^cGM in Figure 1-23. With all elemental probabilities properly recorded, we can begin to compute the conditional probabilities in the tree. For example, look at the node MB where the division into the two distinctions $G$and G^c occurs. The probability of arriving at this point must be the sum of the probabilities of the the two elemental possibilities MBG and MBG^c or $0.032 + 0.112 = 0.144$. Thus the amount of certainty at the point MB to be divided between the distinctions G and G^c is 0.144. In other words, the probability of $MB = \text{P}(MB \mid \mathcal{E}) = 0.144$. We record this in an oval adjacent to the node to show that this is the probability of arriving at that point in the tree.

1.2.9.2.1 Node Probabilities

We call such a probability a node probability; it is the probability that would be recorded as an elemental probability if the tree did not continue to further distinctions. The node probability $\text{P}(MB \mid \mathcal{E}) = 0.144$ would be an elemental probability for the tree of Figure 1-26 if it did not have College Graduate as a third distinction. Similarly we compute the node probabilities $\text{P}(MB^c \mid \mathcal{E}) = 0.408$, $\text{P}(FB \mid \mathcal{E}) = 0.056$, and $\text{P}(FB^c \mid \mathcal{E}) = 0.392$, as shown in Figure 1-26. We repeat the process one stage to the left by computing the probability of the node M as 0.552, the sum of the node probabilities for node MB and node MB^c or $0.144 + 0.408$. A similar computation produces 0.448 for the node probability of the node F. All the node probabilitie[s are] shown in ovals adjacent to the nodes.

1.2.9.2.2 Computing Conditional Probabilities from Node Probabilities

Once we have the node probabilities at all nodes, the conditional probability at any branch is computed by dividing the node probability at the end of the branch by the sum of the node probabilities following all branches of this distinction. We can begin in any part of the tree that we like. For example, to find the conditional probability of B given M, we write P$(B \mid M, \mathcal{E}) = 0.144 \div 0.552 = 0.261$, as shown in the tree. To compute the probability of G given F and B^c, we write P$(G \mid FB^c, \mathcal{E}) = 0.224 \div 0.392 = 0.571$. Of course, for the M-F distinction the conditional probabilities are the node probabilities. We see that it is a simple mechanical procedure to change the order of any probability tree and, consequently, of any relevance diagram. In professional settings, the computations are automatically performed by computers.

1.2.9.2.3 Assessing Relevance

Now that we have the probability assignments for the tree in Figure 1-26, let us examine them in some detail. We note, for example, that P$(G \mid MB, \mathcal{E}) = 0.222$ and P$(G \mid FB, \mathcal{E}) = 0.143$. From either Figure 1-16 or Figure 1-23 we know that P$(G \mid B, \mathcal{E}) = 0.2$. This means that you have assigned the probability of College Graduate given Beer Drinker as 0.2 in the absence of any additional information. However, if you are told, in addition, that the person is a Male, you will increase that probability to 0.222 and if you were told that the person is a Female, you will decrease it to 0.143. This shows the relevance of Sex to College Graduate when the person is known to be a Beer Drinker. We can, thus, readily determine the inferential value of any piece of information.

1.2.10 Computing the Probability of Any Set of Distinctions Given Any Information Represented by Distinctions

We have said that a probability tree is sufficient to compute the probability of any set of distinctions given any other set of distinctions as long as all distinctions are represented in the tree. Let us compute the probability that the person is a college graduate given that he is a male, P$(G \mid M, \mathcal{E})$, from the probability tree of Figure 1-26.

1.2.10.1 Reordering the Tree

We could, of course, reorder the tree in the order College Graduate–Sex–Beer Drinker and simply read the probability P$(G \mid M, \mathcal{E})$ at the second layer. This procedure should now be clear. It will always work, but sometimes smaller computations are adequate.

1.2.10.2 Using the Chain Rule and Selected Node Probabilities

For example, if we wish to work using the tree in Figure 1-26, we have a choice of methods. First we recall from our chain rule discussion that P$(GM \mid \mathcal{E})$ = P$(M \mid \mathcal{E})$P$(G \mid M, \mathcal{E})$, and hence that the probability we seek is given by P$(G \mid M, \mathcal{E})$ = P$(GM \mid \mathcal{E})$/P$(M \mid \mathcal{E})$. From the tree we can compute the probability P($GM \mid \mathcal{E}$) as the sum of P($MBG \mid \mathcal{E}$) and P($MB^cG \mid \mathcal{E}$); or $0.032 + 0.336 = 0.368$; the probability of M, P($M \mid \mathcal{E}$), is 0.552 as shown. Hence P$(G \mid M, \mathcal{E}) = 0.368/0.552 = 0.667$. You have assigned a probability of about two-thirds that if a Male enters the room he will be a College Graduate.

1.2.10.3 Recognizing a Conditioned Probability Tree

Another way to compute this same probability is to consider the portion of the probability tree in Figure 1-26 that follows the branch M to be a probability tree that is already conditioned on the person being a Male. Within this tree we can then ask what is the probability that the person is a College Graduate. This is just the probability of traversing the remaining branches BG or B^cG, or $(0.261)(0.222) + (0.739)(0.824) = 0.667$, the same result.

We may also be interested in the probability that if a Female enters the room she will be a College Graduate, P$(G \mid F, \mathcal{E})$. By any of the methods we have discussed we obtain the result of about 0.52, as you may wish to check. Thus, the probability assignments you have made indicate that in this situation the probability that the next person entering the room will be a College Graduate is higher if the person is Male than if she is Female.

1.2.11 Probability Trees with Several Degrees of Distinction

Now let us examine, in more detail, probability trees with multiple degrees of distinction. Recall the possibility tree of Figure 1-9 where we had three levels of distinction for drinking beer and four levels of distinction for education. We show probability assignments for this possibility tree in Figure 1-27.

Figure 1-27. Probability Tree with Multiple Degrees of Distinction for Two Kinds of Distinction

The probabilities on the first distinction B_1,B_2, B_3 are those from Figure 1-13. You now have to assign probabilities to different amounts of education given different degrees of beer drinking. These are recorded in the second layer of the tree. Note that as the level of beer drinking increases, the probabilities assigned to different levels of education shift to be greater for lower levels. The figure also shows the elemental probabilities obtained by multiplying the branch probabilities.

1.2.11.1 Deducing Probabilities

We can deduce from this tree probabilities you have assigned to different levels of education alone by considering them as collections of elemental possibilities. Thus the probability of at most a grade school education G_1 is the sum of the elemental probabilities of B_1G_1, B_2G_1, and B_3G_1, or $0.07 + 0.04 + 0.03 = 0.14$. When such a calculation is performed for all four degrees of education, we obtain the probability tree of Figure 1-28.

Figure 1-28. Probability Tree for Four Degrees of Distinction for Level of Education

Note that this tree implies a probability of the event College Graduate, $G = G_3 + G_4$, of 0.62. This is slightly different from the probability of the event that you assigned earlier without the detailed thought that went into Figure 1-27. Such discrepancies are to be expected.

1.2.11.2 Multi-degree Relevance

The existence of multiple degrees of distinctions will allow us to refine the definition of relevance we used for simple distinctions. Consider a distinction named A with m degrees A_1, A_2, ..., A_m and a distinction named B with n degrees of distinction B_1, B_2, ..., B_n. Then A is said to be relevant to B if for at least one degree of distinction A_i and at least one degree of distinction $ B_j$,

$$\mathsf{P}(B_j \,|\, A_i, \mathcal{E}) \neq \mathsf{P}(B_j \,|\, \mathcal{E})$$

To show that if A is relevant to B, then B is relevant to A, consider one such pair A_i and $ B_j . $Let\$A_i$ be our earlier simple distinction E, and all other degrees of distinction of A be E^c. Similarly, let $ B_j$ be our earlier simple distinction F, and all other degrees of distinction of B be F^c. Then by our earlier proof since $\mathsf{P}(F \,|\, E, \mathcal{E}) \neq \mathsf{P}(F \,|\, \mathcal{E})$, then $\mathsf{P}(E \,|\, F, \mathcal{E}) \neq \mathsf{P}(E \,|\, \mathcal{E})$. Therefore, $\mathsf{P}(A_i \,|\, B_j, \mathcal{E}) \neq \mathsf{P}(A_i \,|\, \mathcal{E})$, and we have shown that for multiple degrees of distinction, if one distinction is relevant to another, then the second is relevant to the first.

We must note that even if one multi-degree distinction is relevant to another multi-degree distinction, then it is possible that knowing about the occurrence of one degree of the first distinction may tell you nothing about the occurrence of a particular degree of the second distinction. However, this knowledge must tell you something about the occurrence of some degree of the second distinction.

1.2.12 Measures

We could use the probability tree of Figure 1-29 to explore further the subject of measures.

Figure 1-29. Probability Tree with a Measure

We have shown at each endpoint the dollar winnings you will receive from the bets, as originally specified in Figure 1-11. For each elemental possibility we know both the elemental probability and the associated measure value. Thus, the probability of losing $100 is 0.07 and the probability of winning $100 is 0.21. The same measure value may be associated with several different endpoints. For example, you will receive a payoff 0 for 6 different elemental possibilities. The probability of receiving 0 is the sum of the probabilities of these six elemental possibilities or $0.14 + 0.28 + 0.04 + 0.06 + 0.06 + 0.04 = 0.62$.

1.2.12.1 Probability Distributions

Although the tree contains all information we have on the probabilities to be assigned to different measure values, there is a more convenient way to describe the information. We can construct a graph that shows for each possible value of the measure, the probability of achieving that value. We call this graph a probability distribution. For every possible value of the measure m we construct a bar whose height is the probability that that value will be attained. We give this distribution the notation P$(m\,|\,\mathcal{E})$ and call it the probability distribution on m given your background information $\mathcal{E}$. The probability distribution on dollar winnings, d, for this problem, appears as Figure 1-30.

Figure 1-30. Probability Distribution of Dollar Winnings P$(d\,|\,\mathcal{E})$

The probability distribution is a much more compact and informative representation of the probabilities assigned to d than is the tree. We can see just by looking at it that more of your certainty lies to the right of 0 than to the left. This means that you are more likely to win the bet than you are to lose it. However, the probability of 0.62 at 0 shows that most likely result is that no money will change hands.

1.2.12.2 Cumulative Probability Distributions

An alternate representation of the probabilities assigned to a measure is the cumulative probability distribution, which shows the probability that the measure will take on any value less than or equal to a specified number. For any specified number c, the cumulative probability that m will be less than or equal to c is, in our notation, P$(m \le c\,|\,c, \mathcal{E})$.

The cumulative probability distribution P$(d \le c\,|\,c, \mathcal{E})$.on dollar winnings, d, is shown in Figure 1-31.

Figure 1-31. Cumulative Probability Distribution of Dollar Winnings P$(d \le c\,|\,c, \mathcal{E})$

To construct the cumulative distribution, we begin at the left side. For numbers less than -300, there is no chance that the dollar winnings will be less than or equal to such a number; therefore the height of the cumulative distribution is 0. For numbers between -300 and -200, the probability that the winnings will be less than or equal to such a number will be 0.01, just the chance of losing $300. For numbers between -200 and -100 the probability of winnings less than or equal to such a number is 0.03, the sum of the probabilities of losing $200 and of losing $300.

We see that what we are generating is a staircase. Steps occur where there are bars in the probability distribution; the height of each step is the height of the bar. Since the sum of all the bar heights in the probability distribution is 1, eventually the cumulative distribution rises to a height of one, in this case, at the value of +$300. With probability 1, the dollar winnings will be less than or equal to any amount greater than $300. Note that the probability of winning no is represented by the jump from 0.1 to 0.72 in the value of the cumulative probability distribution at 0.

1.2.12.3 Excess Probability Distributions

Sometimes it is more convenient to talk about the probability that the value of a measure will exceed any specified number rather than be less than or equal to it. We call the probability distribution defined in this way an excess probability distribution. For any number, c, the excess probability distribution on a measure m shows the probability that m will be greater than c; in our notation, P$(m > c \mid c, \mathcal{E})$.

Figure 1-32 shows the excess probability distribution P$(d > c \mid c, \mathcal{E})$ on dollar winnings d.

Figure 1-32. Excess Distribution of Dollar Winnings P$(d \geq c \mid c, \mathcal{E})$

The probability that you will win more than any amount less than -$300 is 1. The probability that you will win more than -$300 is 0.99; the probability that you will win more than -$200 is 0.97. Note that what we have constructed is a staircase in the opposite direction. The excess distribution starts at the left at 1 and falls anywhere there is a bar in the probability distribution by an amount equal to the height of that bar.

1.2.12.3.1 Complementarity with Cumulative Distributions

Since d≤c and d>c represent two mutually exclusive and collectively exhaustive possibilities, the sum of their probabilities must be one. This means that if we add together the values of the cumulative probability distribution and the excess probability distribution at any point, the sum must be 1. If you cut out the excess distribution with a pair of scissors, and invert it on top of the cumulative probability distribution the fit would be perfect.

The question of which of the three representations of the probabilistic assignments on a measure is most appropriate will depend on the application.

1.2.12.4 Multiple Measures

Figure 1-33 adds a second measure called car wash winnings, w, to Figure 1-29.

Figure 1-33. Probability Tree with Two Measures

Whenever you lose money you must wash the other person's car; when you win money you get your car washed. The probability distribution of the measure car wash winnings appears in Figure 1-34.

Figure 1-34. Probability Distribution of Car Wash Winnings P$(w \mid \mathcal{E})$

Since you win money whenever you get your car washed, the probability of 0.28 for the value w = 1 is your probability of winning money shown by the height of the excess probability distribution of Figure 1-32 at the value 1.

1.2.12.4.1 Joint Probability Distributions

We can also construct a joint probability distribution on both dollar winnings and car wash winnings as shown in the three dimensional plot of Figure 1-35.

Figure 1-35. Joint Probability Distribution of Dollar Winnings and Car Wash Winnings $\mathrm{P}(d, w \mid \mathcal{E})$

Here for every point in the d, w plane described by dollar winnings and car wash winnings, we construct a bar whose height is the probability, that this combination of values will be attained. Our notation for this joint probability distribution is $\mathrm{P}(d, w \mid \mathcal{E})$. We can also define joint cumulative and joint excess distributions when necessary.

1.2.13 Using the Characterization Language

The characterization language that we have created is simple, but powerful. We shall illustrate its use by example. In each case, think about the problem for a moment before employing our structured analysis. Then you will be able to judge for yourself the contribution analysis can make to the quality of thought.

1.2.13.1 A Card Problem

Problem: A deck of two red cards and two blue cards is fairly shuffled and then the four cards are dealt face down. Two cards are then turned face up. What is the probability that both cards are red?

Most people answer that the probability of both cards being red is one-half, some people say one-quarter. We apply our characterization analysis as follows. First we construct the possibility tree. We create two distinctions. The first distinction is the color of the first card turned up. The first card turned up may be red, R_1, or blue, B_1. The second distinction is the color of the second card turned up. The second card can be red, R_2 or blue, B_2. The events in Figure 1-36 constitute the possibility tree for this situation.

Figure 1-36. Probability Tree for a Card Problem

There are four elemental possibilities, R_1R_2, that both cards turned up will be red, B_1R_2, that the first card turned up will be blue and the second card turned up will be red, etc. Note that every possible result of turning up the cards is described by an elemental possibility. Next we change the possibility tree into a probability tree by assigning probabilities to the possibilities. Since there are two cards of each color the first card turned up is equally likely to be red or blue, we assign probability 1/2 to each of the two possibilities.

Then we assign the conditional probability to the result of the second card turned up given the result of the first card. If the first card turned up was red, then the remaining face down cards are one red card and two blue cards; the probability of a red card is therefore one-third. If the first card turned up is a blue card, there are two red cards and one blue card remaining face down so the probability that the second card is red is now two-thirds. These probabilities are recorded in the tree. We now multiply the branch probabilities to obtain the four elemental probabilities. We see that the probability of the outcome we are interested in, R_1R_2, is 1/2x1/3 or 1/6.

Note that this analysis requires no creative leaps, but only the straightforward application of the concepts we have been discussing.

1.2.13.2 A Coin Problem

Problem: Three coins are placed in a bag. One has two heads, one has two tails, and the third has both a head and a tail. A coin is removed from the bag and tossed. The results is heads. What is the probability that this is the two-headed coin?

We create two kinds of distinction. The first distinction is the type of coin drawn from the bag. It can be a two-headed coin, 2H, a two-tailed coin, 2T, or a normal coin, with a head and a tail, HT. The second distinction is the result of the toss. It can be a head, H, or a tail, T. The possibility tree appears on the left side of Figure 1-37.

Figure 1-37. Probability Tree for a Coin Problem

The six elemental possibilities are that the two-headed coin produced a head, 2H-H, that the two-headed coin produced a tail, 2H-T, etc.

We are now ready to assign probabilities. By the statement of the problem, each type of coin is equally likely to be drawn, so we assign probability 1/3 to each possibility for the first distinction. Then we assign the probability of tossing a head with each type of coin. If the coin is two headed then the probability of throwing a head is 1; if the coin is two-tailed, the probability of tossing a head is 0; if the coin is a normal coin, then the probability of throwing a head is 1/2. We then multiply the branch probabilities to obtain the elemental probabilities. For example, the probability that the coin was two-headed and that it produced a head is 1/3.

What we wish to compute is the probability that the coin was the two-headed coin given that it produced a head. This requires that we reverse the order of distinctions in the tree so that this conditional probability will appear on the reversed tree. The reversed tree appears on the right of Figure 1-37. As usual we begin by copying the elemental probabilities into their appropriate places. We see that there is a total probability of 1/2 following each of the branches H and T and that the probability that the coin was two-headed given that it produced a head is 1/3 ÷ (1/3+0+1/6) or 2/3. Given that a head was produced, there is no chance of a two-tailed coin and a 1/3 chance of a normal coin. Note that our analysis provides the answer not only to the question as posed, but to any other question that can be asked about this problem.

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2 The Rules of Actional Thought

Ronald A. Howard

Section 2: The Rules of Actional Thought

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2.1 Rules of Actional Thought

To achieve your goal of making a reasoned decision, you will want to conform your actional thought to rules that you have carefully considered and wish to follow in decision-making.

2.1.1 Considerations

The most general considerations in choosing any set of rules to guide logical behavior are, first, that the rules be correct when applied to simple situations, and, second, that we see no reason why they should break down when extended to the larger situations in which they will actually be used. Thus, we test the rules of arithmetic by checking that 3 plus 4 is indeed 7 by the test of counting. As we extend our use of arithmetic to larger and larger problems we note that the extension is seamless, that there is no apparent point at which dealing with larger or more numbers causes the process to break down.

These same considerations apply to choosing rules for actional thought. Our first goal is that the rules should be transparent, describing what you would actually do in the simple situation to which they apply. They should be so transparent that if you tried to explain what you were doing to an onlooker, you would be treated with impatience, and the interjection, "Of course." However, as we combine these simple rules, you will obtain power over complex decisions just as the simple rule of arithmetic gives you power over numbers. When you face an opaque decision situation, with many alternatives, several sources of uncertainty, and complex preferences, you will be able to combine these transparent rules to transform the opaque decision into a transparent one. You will thereby attain your goal of achieving clarity of acti

2.1.2 The Decision Situation

The rules of thought apply to a particular decision situation. Consider yourself at an epoch of time separating the past from the future. You face choices among different alternatives, different allocations of resources. Each alternative will produce a deal. A deal is composed of various future prospects whose realization may be uncertain. These prospects are not simply future happenings, but your future life with those happenings. Thus, if you were given a chance to flip a coin for $100, one prospect is not simply described by winning $100, but rather your whole future life as it would appear with the winning of $100. While this clarification may not be essential when the prize is $100, it becomes crucial for most people when the prize is $100,000. We therefore think of a decision as a choice between uncertain futures, as a choice of deals with uncertain prospects.

Your actions and future actions determine the deal you are considering; uncertainty about the prospects of the deal originates in events over which you do not have control. Prospects may possess many attributes for which you have a preference; these attributes may be distributed over time.

All these thoughts about a decision occur at this particular epoch. All notions of consistency apply to your thoughts at this epoch and not necessarily to your preferences over time. Just because you prefer chocolate ice cream to vanilla ice cream today does not mean you will prefer it tomorrow. As Emerson said "a foolish consistency is the hobgoblin of simple minds." The consistency we seek is the consistency of our rules of thought, not the foolish consistency of constancy of preference. However, there is nothing in the rules to prevent you from being as consistent in your preferences over time as you like. In fact, most of us will prefer to be relatively consistent in our preferences over time so that we are not buying tomorrow what we gave away today.

2.1.3 The Rules

If you adopt the following five rules, you can achieve high quality actional thought.

2.1.3.1 The Probability Rule

The probability rule requires that you use possibilities and probabilities to provide the distinctions and information that characterize deals. We can think of a deal in principle as a probability tree, perhaps with several measures specified at the endpoints. Since we have described characterization at length, this will require no further discussion.

2.1.3.2 The Order Rule

The order rule requires that you can arrange prospects in a list in descending order of your preference from best to worst. If you prefer prospect A to prospect B (written A>B), then A appears higher in the list than B. Two prospects to which you are indifferent can appear at the same level in a list, a condition described by A=B.

2.1.3.2.1 Creating Requisite Distinctions

If you have trouble in ordering prospects in such a list, it may be because you have not isolated sources of uncertainty in the prospects. For example, if you have difficulty saying whether you would prefer your life with the gift of a new stereo to your life with a gift of a dog, it may be because you do not know how much care a dog requires. You might prefer the stereo to the dog if the dog requires ten hours of care per week, but the dog to the stereo if the dog required only one hour of care per week. By introducing the additional kind of distinction, hours of care for the dog, with sufficient degrees to describe the possibilities, you could then arrange your prospects in order of preference in the list. The rule allows you to introduce as many kinds and degrees of distinction as necessary to enable you to rank prospects. Your acceptance of the rule affirms your belief that you can always create the distinctions necessary to express a preference between prospects.

2.1.3.2.2 Consistency of Order

The order rule prohibits a prospect's appearing at two different levels in the list. This is clear for two prospects, for if A is preferred to B, and hence is higher in the list, then B cannot be higher in the list than A. The rule also requires that if A is above B, A>B, and B is above C, B>C then A above C, A>C. If A is preferred to B and B is preferred to C, then A is preferred to C. If you are tempted to violate this requirement it is usually because you are thinking of different attributes in making the comparison of prospects.

For example, suppose you are given a choice of three cars. The Ajax, the Rex, and the Lap. On reflection you find that you prefer the Ajax to the Rex because of its acceleration; you prefer the Rex to the Lap because of its reliability; and you prefer the Lap to the Ajax because of its luxury. It appears as if you are having trouble in creating an ordered list of preferences. But the question is, all attributes considered, if one of these cars was waiting for you as a free gift in the parking lot, which would you choose? Whichever it is should be highest on the list. Then if that car were not available, which of the remaining two would you choose? That car would rank below the first on the list; finally the remaining car would be below the other two. Remember that we are ranking prospects, where each is your whole life as described by an elemental possibility and its associated measures.

2.1.3.2.3 The Money Pump Argument

There is a major disadvantage of violating the order rule. Suppose you have your original confused preferences about the three cars and that you have just been given a Lap. Then someone could come up to you and say, "I know that you prefer the Rex to the Lap becaus its reliability. If you give me some money, I will exchange your Lap for a Rex." You say "fine" and make the exchange. Then the person reappears and says "I also know that you prefer an Ajax to a Rex because of its acceleration, and so if you give me a little more money, I will exchange your Rex for an Ajax." You say "agreed" and hand over some more money. But we are not through. He comes back and says, "I know you prefer the Lap to an Ajax because of its Luxury. Give me some more money and I will exchange your Ajax for a Lap." You say "OK," and hand over some more money. Now you are back where you started having dispersed money three times. There is nothing to prevent this person from coming back and taking you around the cycle again. This is sometimes called being made a "money pump." Anyone who violates the order rule is vulnerable to a money pumper.

Usually there will be no difficulty in satisfying the order rule. If someone asks you if you would rather have \$100, \$50, or nothing, you will not hesitate to assign your preference. Even with more complex descriptions of prospects, the choice will usually be simple. For example, suppose that your local daily newspaper is sponsoring a drawing to increase circulation. Three of the prizes in the drawing are A, a two-week trip to Tahiti for two people, all expenses paid; B, a \$500 gift certificate at the largest department store in town; and C, a free one-year subscription to the newspaper. Most of us would have little trouble in establishing the preference order A>B>C, specifying that A is above B and B is above C in our preferentially ordered list of prospects. However, any ordering is allowed.

2.1.3.3 The Equivalence Rule

Suppose that you prefer prospect A to prospect to B, and prospect B to prospect C: A>B>C. This means that A is above B and B is above C in the preference list. Then the equivalence rule requires that you can specify a number p so that you will be indifferent between receiving prospect B for sure and receiving a deal with probability p of prospect A and probability $1-p$ of prospect C. Figure 2-1 shows the requirement of the equivalence rule graphically.

Figure 2-1. The Equivalence Rule

You have adjusted the probability p in the probability tree on the right of the figure until you are indifferent between receiving the deal described by that tree and the deal described by the tree to the left of the figure; namely, receiving B with probability 1. You are the sole arbiter on what the setting of p should be to achieve your indifference. The number p can be any number between 0 and 1, but cannot be exactly equal to either. If you set p to 1, you would be saying that you were indifferent between A and B, which is not so. If you set p to 0 you would be saying that you were indifferent between B and C, which is also not the case.

2.1.3.3.1 Preference Probability

When you have found the p that makes you indifferent, we call it the preference probability [illegible] this situation. We call it a preference probability rather than simply a probability because it [illegible] the probability assignment on any event whose occurrence could be foretold by a clairvoyant.

2.1.3.3.2 Thinking of the Probability Wheel

You can think of the equivalence rule as requiring you to create a hypothetical deal involving the prospects A and C that you find equally attractive to the prospect B. One way to think of this is to imagine using the probability wheel to describe the deal, with one color corresponding to winning prospect A and the other to winning C. If, for the particular wheel setting in front of you, you prefer the wheel deal to the prospect B, then you must decrease the color corresponding to A. If you prefer prospect B, then you must increase the fraction of the wheel corresponding to A. When you are indifferent the fraction of the wheel corresponding to A is the number p that you seek for the equivalence rule.

2.1.3.3.3 Certain Equivalent

For the p specified by the equivalence rule, B is called the certain equivalent of the A-C deal.

To illustrate these concepts, let us return to the newspaper drawing. Suppose that after reflection and with the use of the wheel, you are just indifferent between receiving on the one hand, the $500 gift certificate at the major department store, and on the other a deal consisting of a 15 percent chance of the trip to Tahiti and an 85 percent chance of a free one-year subscription to the newspaper. Then we would say that your preference probability for these three ordered prizes is 0.15, and that the $500 gift certificate is a certain equivalent of a deal consisting of a 0.15 chance of a two-week trip to Tahiti and a 0.85 chance of a free one-year subscription to the newspaper.

2.1.3.4 The Substitution Rule

The substitution rule requires that you be indifferent between receiving any prospect and any deal for which you have established that prospect as a certain equivalent. This means that either the prospect or the equivalent deal may be substituted for each other at this epoch. For example, in the case of the newspaper drawing, suppose that you received a call from the paper informing you that you had won the $500 gift certificate. The caller informs you that if you wish, you can forgo the gift certificate and be placed in a drawing for the two-week trip to Tahiti with, as a consolation prize, a free one-year subscription to the paper. If when you inquire about the rules for the drawing, you assign a 0.15 chance of winning the Tahiti trip, then you must be indifferent between exchanging the gift certificate for the drawing and not exchanging.

2.1.3.4.1 Preference Probabilities Treated as Probabilities

This means that when faced with an actual uncertain deal on A and C with a probability of winning A equal to your preference probability, you will be indifferent to receiving that deal and B for sure. In other words, your preference probabilities will describe your actions when they are equal to probabilities you assign. In short, it means that you will treat your preference probabilities as probabilities in analyzing decision situations.

2.1.3.5 The Choice Rule

Suppose that you prefer prospect D to prospect E, $D > E$, which means that D is above E in your preference list. Suppose further that you have a choice of two deals, each of which could yield either D or E, and that you have assigned a probability of winning D in each deal. The choice rule requires that you must prefer the deal with the higher probability of winning D. This is the last rule, and the only rule that specifies the action that you must take. The rule says that you must choose the deal with the higher probability of the prospect you like better.

This rule is so transparent that if you announced you were violating it, people would think you were joking. Suppose you prefer $100 to nothing and you were given a choice between flipping a coin to win $100, to which you assign a probability of 1/2 of winning, and drawing a heart from a deck of cards for the $100, to which you assign a probability 1/4 of winning. If you told a friend that you had chosen the card game, he would think that he had misunderstood you. Notice that someone who is a gambler and who enjoys risk will still prefer the coin deal because he believes he has a better chance of winning.

In the newspaper drawing example, the choice rule requires that if you are given a choice of participating in different drawings with possible prizes of the Tahiti trip or the free one-year subscription to the newspaper, then you must prefer the drawing with the best chance of winning the Tahiti trip, presumably the one with the fewest participants.

The choice rule says nothing about your preferences, but only that you must follow them. Even masochists can follow the choice rule. If you prefer being beaten up to not being beaten up, and you see two deals with different probabilities of your being beaten up, then you must prefer the deal with the greater chance of the beating. *De gustibus non disputandum est*—there is no arguing about taste.

2.1.4 What the Rules Require

We shall soon demonstrate that any decision may be made by repeated applications of these five rules. We now point out some general properties of this normative form of decision-making. We note for example, that the choice between two alternatives depends only on the possible prospects of these alternatives and not on the prospects of other alternatives. Your preferences are on the future prospects you receive, not on what you might have gotten. At this epoch, the removal of an alternative that you did not choose cannot change the order of your preferences among those that remain. These observations may seem strange in the abstract, but they are no more than formalized common sense.

Consider the story of the man who entered an ice cream parlor and asked what flavors were available. The server replied, "vanilla, chocolate, and strawberry." The customer said, "I'll h vanilla." A moment later the server returns saying, "I'm sorry we have no strawberry." The customer says, "In that case, I'll have chocolate." That this story seems to be a joke shows the extent to which we have internalized the rules of thought.

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3 The Party Problem

Ronald A. Howard

Section 3: The Party Problem: Basic Form

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3.1 Using the Rules—The Party Problem

The best way to appreciate the rules of thought is to see them in action in a decision process. We shall use a problem called The Party Problem as an extensive example in which to demonstrate many principles of decision analysis. The Party Problem was developed to be almost the simplest type of decision problem to consider, and yet it is demanding enough to challenge the quality of our thought in contemplating it.

3.1.1 Statement of the Problem

Suppose that my daughter Kim has invited her teen-age friends to a party. However, she seems to be unsettled and so I ask what is the matter. She replies, "I just don't know where to have this party I'm planning." I say, "Since you're perplexed about a decision, perhaps I can help." I ask her what alternatives she is considering, and she says, "There are three locations where I can have my party. I can have it outdoors on the lawn, indoors in the family room, or I can have it on the porch, which, as you know, is covered, but open to the weather on the side." I say, "So what?" She says, "Well I don't know what the weather is going to be like. If it rains and I have the party outdoors, we could be miserable, but if it is sunny and I have it anywhere else , I'm going to wish we were outdoors."

3.1.2 The Decision Tree

At this point I say, "Let us diagram your decision," and I sketch Figure 3-1.

Figure 3-1. The Decision Tree

I draw a square decision node and label the three alternatives emanating from it as O, P, and I, corresponding to the party locations being outdoors, on the porch, or indoors. Then for each alternative I show two possibilities for the weather, sunshine, S, and rain, R. She says, "I'm not sure I follow exactly what you are doing." So, for the sake of efficiency, I give her the notes that you have been reading to review and then resume the discussion a few hours later.

3.1.2.1 The Clarity Test

She begins, "I see what you mean now by the two possibilities for the weather, but what exactly do you mean by sunshine and rain." I say, "That is for you as a decision-maker to define in a way that meets the standards of the clarity test and is also useful in ranking your prospects." We then enter into a twenty-minute discussion of the most appropriate definition of sunshine and its complement rain, which we will not recount here. At the end of this time, however, she is satisfied that she has made this distinction clearly and in the most useful way for her purposes. We shall have had to deal with such issues as cloudiness, precipitation, intermittent clearing with precipitation, etc. in arriving at this simple, dichotomous distinction.

3.1.2.2 Acceptance of the Rules

I then say, "By the way, do you agree to subscribe to the rules of thought that you have read." She says, "Yes, they seem most convincing to me. I shall be happy to follow them." I reply that she has already begun to satisfy the Probability Rule by her careful construction of the decision tree of Figure 3-1.

3.1.3 The Order Rule

I now point out to her that the decision tree she has constructed has created six prospects, combinations of her actions and the possibilities that may arise from them. These six prospects are: having an outdoor party in the sunshine, O-S, having an outdoor party in the rain, O-R, having a party on the porch in the sunshine, P-S, etc. I then ask her which is the best of all prospects. She replies, "Having an outdoor party in the sunshine, O-S." "And which is the worst?" I ask. She replies, "An outdoor party in the rain, O-R. We would be sodden, but we could still go to a movie or something."

I then ask her to make a list of the six prospects with the best prospect, outdoors in the sunshine, at the top and the worst prospect, outdoors in the rain, at the bottom. She constructs the ranking shown in Figure 3-2.

Figure 3-2. The Preferentially Ordered List of Prospects

At the top is O-S, labeled B as the best prospect; at the bottom is O-R, labeled W for the worst prospect. The other four prospects in order of preference are P-S, I-R, I-S, and P-R. In reviewing the list we note that she has ranked an indoor party in the rain, I-R, higher than an indoor party in the sunshine, I-S, and we ask her if this really reflects her preference. She says, "Yes, for an indoor party in the rain could still be quite nice, but an indoor party in the sunshine might well make people wish they were outdoors." Just to be sure, I ask her if there are any two prospects on this list to which she is, in fact, indifferent. She replies, "No, I like each one better than any below it." I mention that she has now satisfied the Order Rule.

3.1.4 The Equivalence Rule

Now I remind her of the Equivalence Rule. I say, "You have said that outdoors-in-the-sunshine is the best prospect, that outdoors-in-the-rain is the worst prospect, and that, for example, that porch-in-the-sunshine lies between these two in your preference. Correct?" She says, "Yes that's right." Then I say, "In accordance with the Equivalence Rule, we should be able to offer you some chance of the best prospect versus the worst prospect that will make you just indifferent to having a party on the porch in the sunshine for sure." While I am saying this I set up the probability wheel, and tell her that if one color appears the wizard will magically give her an outdoor party in the sunshine, but that if the other color appears he will will magically give her a outdoor party in the rain. I ask her if she would be just indifferent between spinning the wheel on the one hand or receiving a party on the porch in the sunshine, also provided by the wizard, for sure. She says that in fact a porch party in the sunshine could be a very nice party so she would have to have a probability much closer to one than to zero of winning the the best party before she would be indifferent. After adjusting the colors she establishes that if she had a 95 percent chance of an outdoor party in the sunshine and a 5 percent chance of an outdoor party in the rain, she would be just indifferent between this deal and a sure party on the porch with sunshine. I remind her that this 0.95 is her preference probability between best and worst corresponding to the prospect P-S; and that P-S is her certain equivalent for a 95 percent chance of the best as opposed to the worst prospect.

I then ask her to repeat this process for all prospects and she produces Figure 3-3.

Figure 3-3. Creation of Best and Worst Deals Equivalent to Each Prospect

For completeness we have shown that the outdoor-in-the-sunshine party is equivalent to probability 1 of the best prospect and the outdoor-in-the-rain party is equivalent to probability 1 of the worst prospect. As we go down her preference list on prospects, the preference probability on best versus worst falls. It is 0.95 for P-S, 0.67 for I-R, 0.57 for R-S and 0.32 for P-R. Note that these preference probabilities give us some idea of the strength of her preference for the various prospects.

3.1.5 The Probability Rule

At this point I might say that we will leave the issue of her preferences for the moment and return to the issue of information. I observe that although we have completed the structure of the decision tree in Figure 3-1, we have not yet supplied her probability assignment on the event Sunshine. We then begin a discussion on probability assignment, and I ask her to review all the information she knows or can readily obtain about tomorrow's weather. She listens to the television, reads the newspaper, looks at the sky, and finally sits down to have her probability encoded by the processes we have discussed. She assigns a probability of 0.4 to sunshine and consequently a probability 0.6 to rain.

The decision tree with these probabilities appears as Figure 3-4.

Figure 3-4. Decision Tree with Probabilities

We now proceed to combine the alternatives and information for this figure with the preference information of Figure 3-3 to demonstrate the right decision for her. I can assure her that as long as she follows the rules of thought, from this point on the best alternative for her is just a matter of logic.

3.1.6 The Substitution Rule

I remind her of the substitution rule that allows me to substitute for any prospect a deal involving the best and worst prospects for which it is the certain equivalent. I perform this operation in Figure 3-5.

Figure 3-5. Substitution of Equivalent Deals for Each Prospect

Here for each prospect of the decision tree I have substituted the equivalent deal from Figure 3-3. Thus, for example, if she picks the porch alternative and the weather is sunny, she will be in a situation that she regards as equivalent to one with a 95 percent chance of the best prospect, O-S, and a 5 percent chance of the worst prospect O-R.

When we examine Figure 3-5 we note that for each alternative O, P, and I, there are only two possible prospects, the best prospect, B, and the worst prospect, W. Since we have agreed in the substitution rule that there is no difference between probabilities and preference probabilities, we can compute the preference probability of achieving the best prospect for each alternative by multiplying together the conditional probabilities on the branches of each path that leads to the best prospect and then adding over all paths. Thus, in Figure 3-5, if the Outdoor alternative is selected, there is a 0.4 chance of Sunshine and a 1 chance of best prospect given Sunshine; there is a 0.6 chance of Rain and a 0 chance of best prospect given Rain. The chance of best prospect given the Outdoor alternative is therefore 0.4x1+0.6x0=0.4. For the Porch alternative, the chance of obtaining the best prospect is 0.4x0.95+0.6x0.32 or 0.57. For the Indoor alternative the chance of obtaining the best prospect is 0.4x0.57+0.6x0.67 or 0.63. These results are recorded in the simplified tree of Figure 3-6.

Figure 3-6. Alternatives Represented as Probabilities of Best Prospect

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Note that the Outdoor, Porch, and Indoor alternatives produce preference probabilities 0.4, 0.57, and 0.63 of the best prospect as opposed to the worst.

3.1.7 The Choice Rule

We now invoke the choice rule and observe that since she prefers the best prospect to the worst prospect, she must select the alternative with the highest preference probability, the highest equivalent probability of achieving the best prospect. This is the Indoor alternative for the party. We note that it is equivalent to a 63 percent chance of the best prospect and a 37 percent chance of the worst prospect.

In interpreting this result for her, we point out that although her best alternative, the Indoor alternative, is equivalent to a 63 percent chance of the best versus the worst prospect, in fact, there is no possibility for her to achieve either the best or the worst prospect by following this alternative. There is no real world event with a probability 0.63 in this problem. We have computed her preference probability for each alternative, not the probability of an event. What she has established is that having the party indoors is equivalent to a 63 percent chance of the best versus the worst prospect, and that this is a better chance of the best versus worst prospect than she could obtain with any other alternative. If at this point the wizard were standing by to produce either an outdoor party in the sunshine or an outdoor party in the rain, she would be just indifferent to spinning a probability wheel that would give him the first instruction with probability 0.63 and the second instruction with probability 0.37.

3.1.8 Expected Value

By defining a new operation, we can save ourselves one step in deriving the best alternative. Suppose we constructed the decision tree in the form of Figure 3-7, where we have shown as a measure the preference probability to be associated with each prospect as given by Figure 3-3.

Figure 3-7. Preference Probability for Each Prospect as a Measure

We note that if for each alternative we multiplied for each elemental possibility its elemental probability and the value of the measure and then summed over all elemental possibilities, we would obtain the preference probability for the alternative as given by Figure 3-6. These preference probabilities are shown underlined in the decision tree of Figure 3-7.

Consider a probability tree with a measure. We compute for each elemental possibility the product of its elemental probability and the associated value of the measure. Then we sum over all elemental possibilities. We call the number we have computed the expected value of the measure. With the expectation operation defined, we can now say that the preference probability for any alternative is the expected value of the preference probabilities for each prospect. We shall have other uses for the e- operation as we proceed.

3.1.9 Money

The procedure we have described can be used to analyze any decision. However, it is often useful to introduce money as a measure of the desirability of prospects. There are two main reasons for introducing money. The main one is that we have a lifetime of experience in using a monetary measure in valuation. We are often much more comfortable in the consistency and reasonableness of our assessments when we use the concept of money in the process.

A second and equally important reason for using money or some other fungible resource in value assessments is that we can compute the value of information only when we have introduced a resource in which to measure it. In the party problem as we have formulated it up to this point there is no way to discuss the value of information on the weather.

Some people become uncomfortable when money is introduced into a decision situation that seems distinctly non-monetary, as if discussing money were demeaning. We might recall that the famous injunction does not say "money is the root of all evil" but rather that "the love of money is the root of all evil." Money is just money, a measure of worldly resources.

3.1.9.1 Bring On the Wizard

The way we introduce money or an equivalent resource like time is to ask the decision-maker for his willingness to pay in terms of this resource for changes from one prospect to another. To free the decision-maker's mind from questions about the likelihood of such changes, we employ the wizard to effect the changes and ask only about the willingness to pay the wizard for his services.

3.1.10 Kim's Values for Party Prospects

We illustrate the procedure in the party problem by determining Kim's value in dollars for each of the party prospects. For simplicity we shall assume that she is just indifferent between having an outdoor party in the rain and having no party at all. She says that if she ended up with an outdoor party in the rain they would probably call the whole thing off and go to the movies and the whole notion of the party would simply have served as a way to get together.

3.1.10.1 Willingness-to-Pay to Change Prospects

We then ask, suppose you were in the position of having arranged for an outdoor party and it is now just about to rain. The wizard appears and offers to make the event Sunshine happen. How much would you be willing to pay for this service. After some reflection she answers, "$100." We confirm that she would not pay more than $100 for the service and that she would pay any amount less than $100. What she means by this statement is that she is indifferent between having an outdoor party in the rain with her present bank account and having an outdoor party in the sunshine with $100 less in her bank account.

I then suggest that the wizard may not offer to make it sunny, but only to move her party to the porch while the rain still occurs. She considers this service much less valuable and will pay only $20 for it. But if when she was on the porch in the rain he further offered to make it sunny she would pay an additional $70 for that service. We confirm that this means that if he offered to move her from an outdoor party in the rain to a porch party in the sunshine she would pay a total of $90, and she replies, "Yes."

Next I ask how much she would pay to have the wizard transform her outdoor party in the rain to an indoor party in the rain. She says, "$50." I then ask how much more she would pay to continue the change to an indoor party in the sunshine. She says, "You have forgotten that I prefer an indoor party in the rain to an indoor party in the sunshine. In fact, I would pay him $10 more not to change the weather from rainy to sunny given that I was committed to an indoor party." I say, "You mean you would pay only $40 to go from an outdoor party in the rain to an indoor party in the sunshine." And she says "Yes."

3.1.10.2 Prospect Dollar Values

These valuations are recorded in Figure 3-8.

Figure 3-8. Dollar Value of Each Prospect as a Measure

The amounts Kim will pay to change any prospect into any other prospect is the difference in the values of the prospects. We note that the higher the value of a prospect, the higher its position in the list in the Figure 3-2. Figure 3-9 shows the preferentially ordered list of prospects with columns of preference probabilities and dollar values.

Figure 3-9. Preference Probability and Dollar Value Measures for Each Prospect

Prospect	Preference Probability	Dollar Value
Outdoors, Sunshine	1	$100
Porch, Sunshine	0.95	$90
Indoors, Rain	0.67	$50
Indoors, Sunshine	0.57	$40
Porch, Rain	0.32	$20
Outdoors, Rain	0	$0

The higher the preference probability, the higher the dollar value. For these six prospects we can plot preference probability versus dollar value as shown in Figure 3-10.

Figure 3-10. Plot of Preference Probabilities versus Dollar Values

We can think of this plot as a scaling from dollars to preference probability. If we had had this plot for Kim for the dollar values she assigned for the prospects, then we could have used it to obtain the preference probabilities for Figures 3-5 or 3-7 and thereby found the preference probability for each alternative.

3.1.10.3 The Utility function

The form of Figure 3-10 suggests that there is a relationship between Kim's preference probabilities and her value in dollars. Note that for purposes of making the decision, the vertical scale of Figure 3-10 is irrelevant. If we had multiplied all the preference probabilities by 10 and then carried out the expectation operation of Figure 3-7, the preference probabilities for each alternative would merely be multiplied by 10. The one that was largest before would still be largest. The same would be true if we added or subtracted any number from all the preferences probabilities in the plot. The same number would merely be added or subtracted to the preference probabilities for each alternative; the alternative with the largest preference probability would be unchanged. While such transformations would no longer permit the interpretation of the results as a preference probabilities, they would lead to the selection of the correct alternative. It is sometimes numerically convenient to perform such a transformation and to forgo the preference probability interpretation. If at any time it is desired to interpret the results as a preference probability, it is merely a matter of defining a best and a worst prize and of rescaling.

3.1.10.3.1 Kim's Utility function

Kim's preference probability plot of Figure 3-10 has been smoothed and replotted as Figure 3-11.

Figure 3-11. Smoothed Plot of Preference Probabilities versus Dollar Values: Kim's U-Curve

Notice that the plot extends beyond the value 1 and below the value 0; we have forgone the interpretation as preference probability outside the range from $0 to $100. We call a curve like Figure 3-11 a utility function and the value it produces at any point the utility .

The utility function as plotted is adequate for the following discussion; however, if you wish to confirm our calculations, you should know that Kim's particular utility function has the form

$$u(x) = \frac{4}{3}\left[1 - \left(\frac{1}{2}\right)^{x/50}\right]$$

This means that to find the utility corresponding to any dollar measure x, we raise $\frac{1}{2}$ to the power x divided by 50, subtract the result from 1, and then multiply by $\frac{4}{3}$.

Once we have Kim's utility function and her dollar valuation on prospects, we use the utility function to obtain the utility of each prospect, and then find the best alternative for her by finding the alternative with the highest expected value of utility . It is clear that this procedure will choose for her the Indoor alternative in the party problem.

3.1.11 Kim's Certain Equivalent of a 50-50 Chance of $100

Kim's utility function may be used to help her make other decisions having nothing to do with the party. We can use Kim's utility function to obtain her certain equivalent for any monetary deal that is within the scope of the utility function.

For example, suppose she is offered a deal which she assesses as a 50-50 chance of winning $100 as opposed to nothing. What should her certain equivalent in dollars for this deal be? We could answer this question without consulting her once we have her utility function from Figure 3-11.

We know that her certain equivalent will be the amount of money she would have to receive for sure to be indifferent between that prospect and the deal with a 50 percent chance of winning $100 and a 50 percent chance of winning nothing. We know that if she is to be indifferent, the certain equivalent and the deal must have the same preference probability or utility . The utility of the deal is the expected value of the utilities of its prizes. The calculation is shown in Figure 3-12.

Figure 3-12. Kim's Certain Equivalent for a 50-50 Chance of $100

According to Figure 3-11, the prize of $100 has a utility of 1; the prize of $0 has a utility of 0. The expected value of these utilities is $0.5 \times 1 + 0.5 \times 0 = 0.5$, the utility of the deal. If she is to be indifferent to the certain equivalent, it must also have a utility of 0.5. By referring to the utility function of Figure 3-11, we find that a utility of 0.5 corresponds to a dollar value of about $34. Consequently, the sum of $34 and the deal both have the same utility of 0.5. This means that $34 is Kim's certain equivalent for the 50-50 chance at $100.

3.1.11.1 Graphical Interpretation

When a deal has only two possible monetary payoffs, we can find its certain equivalent from the utility function by a simple construction that implements this set of calculations. First we draw a straight line between the two points on the utility function corresponding to the utilities of the payoffs. Then we find the point on this line such that the distance from the lower utility to this point divided by the total length of the line is equal to the probability of the larger payoff. Finally, we move horizontally to the left from this point until we meet the utility function and drop down to read the certain equivalent.

Figure 3-13 illustrates the procedure for the 50-50 chance at $100.

Figure 3-13. Graphical Construction to Find Certain Equivalent

The dashed straight line connects the points on the utility function corresponding to 0 and 100 dollars. Since the probability of winning the higher amount, $100, is 1/2, starting from the lower point, we proceed 1/2 the way along the line and locate the point shown as a large dot. The height of this point is the utility of the deal. Then we move left from this point until we encounter the utility function and read off the corresponding certain equivalent, $34, on the dollar scale.

3.1.12 Kim's Certain Equivalents of Party Locations

We can also use Kim's utility function to obtain her dollar certain equivalent for the three party locations. We know from Figures 3-6 or 3-7 that the Outdoor, Porch and Indoor alternatives correspond to preference probabilities or utilities of 0.40, 0.57, and 0.63. By referring to the utility function of Figure 3-11, we find that the dollar certain equivalents corresponding to these utilities are $26, $40, and $46. You may also wish to obtain these certain equivalents by the graphical construction we just discussed.

The certain equivalents are recorded in boxes adjacent to each alternative in the tree of Figure 3-14.

Figure 3-14. Kim's Certain Equivalents for Party Problem Alternatives

Kim would be just indifferent between following the Outdoor alternative and receiving $26; to following the Porch alternative and receiving $40; and to following the Indoor alternative and receiving $46. These certain equivalents give us a feeling for her strength of preference among the three alternatives.

3.1.12.1 Choosing by Certain Equivalent

Since Kim will always prefer more money at least as much as less (she can always give it away), Kim's utility function can never decrease as dollars increase. This means that the higher her preference probability or utility for an alternative, the higher her certain equivalent for that alternative. Consequently she can determine the best alternative to follow either by choosing the one with the highest preference probability—utility —or by choosing the alternative with the highest certain equivalent.

3.1.13 Valuing Clairvoyance

Suppose that at this juncture in Kim's saga of the party problem the clairvoyant arrived with an offer of help. Recall that the clairvoyant is competent and trustworthy—he knows the future weather and he will tell the truth if we engage him. However, unlike the wizard, he cannot change anything. The clairvoyant offers to tell whether the event sunshine S as Kim has defined it will occur. Note that he does not describe the weather at the time of the party in some general sense, but only in terms of an event that meets the clarity test.

The good news is that the clairvoyant has arrived; the bad news is that he wants to be paid. His fee for his service is $15. Should Kim hire him?

The question is whether she would be better off by doing the best she can without him or by hiring him for $15 and then deciding on the party location with the knowledge of whether S will occur.

We represent Kim's decision on buying clairvoyance at a price of $15 by the decision tree of Figure 3-15.

Figure 3-15. Determining the Value of Clairvoyance on S/R at a Cost of $15

The upper part of the tree represents the situation where she does not buy clairvoyance. She then faces the same choices for party location and the same consequences shown in Figure 3-14. As we have shown, her best alternative in this case is to have the party indoors with a utility of 0.63 and a certain equivalent of $46.

The lower portion of the tree represents the clairvoyance deal at a price of $15. If she takes this deal, the first thing that will happen after committing herself to the payment of $15, is the report by the clairvoyant on what will happen. We let "S" represent the event that the clairvoyant reports that the event S will occur and "R" be his report that it will not. We are uncertain about what he will say because we are uncertain about the weather. Since he will report "S" when and only when S will occur, "S" and S represent the same distinction and must be assigned the same probability; in this case, 0.4, with 0.6 probability assigned to "R". Once the clairvoyant's report is known, Kim will be able to choose a party location, Outdoors, Porch, or Indoors. Then the weather will actually turn out to be sunny or rainy, S or R, and at the end of the tree we show the measure she has attached to each of these prospects. The dollar measure of each prospect is computed by subtracting the $15 that must be paid to the clairvoyant from the measures that we have been using for each prospect. Thus, for example, if she has an outdoor party in the sunshine her value is not $100 as before, but now $85 because of the clairvoyant's payment. All the dollar values at the end of the tree have been reduced by $15 to reflect this commitment.

We have now completed the structure of the tree and the assignments it requires except for noting that when the clairvoyant says "S", then of course the weather will be S with probability 1, and that when he says "R", then it will be R with probability 1. Thus when the clairvoyant reports S, Kim is looking forward to Sunshine and must select between party locations that will pay her the equivalent of $85, $75, and $25 for the locations Outdoors, Porch, and Indoors. Clearly she will choose to have her party Outdoors in this case and obtain the payoff $85. This is shown by the arrow on the Outdoor branch of the location decision following the report "S". If the clairvoyant reports "R", then she faces rainy weather and must choose among the three locations which will then pay –$15, $5, and $35. Again, she will choose the one that gives her the highest payoff, namely to have the party Indoors.

We thus see that the alternative of purchasing clairvoyance at a price of $15 reduces to selecting a deal with a 0.4 chance of an $85 payoff and a 0.6 chance of a $35 payoff. From the utility function of Figure 3-11, the utilities of $85 and $35 are 0.92 and 0.51 as shown adjacent to the "S" and "R" branches. The utility of buying the clairvoyance is then $0.4 \times 0.92 + 0.6 \times 0.51$ or 0.674. From the utility function we find that the certain equivalent of this alternative is $51.

Thus we have determined that buying the clairvoyance at this price is a better deal for Kim than her original deal without it, for the utility of 0.674 is higher than the utility of 0.63 or equivalently, the certain equivalent of the clairvoyant's deal is $51 compared to the certain equivalent of $46 from proceeding without clairvoyance. Kim would be well advised to buy the clairvoyance at a price of $15.

Notice how the clairvoyant's report will affect Kim's actions. If he reports that the weather will be rainy, then she will have an indoor party just as she would have had if she had never consulted him. If, on the other hand, the clairvoyant's report is that the weather will be sunny, then she will change from an indoor party to an outdoor party. The value of the clairvoyant's information comes from the possibility that the information may affect her decision. In this case there is a 40 percent chance that she will change her course of action as the result of engaging the clairvoyant.

3.1.14 The Value of Clairvoyance

We have found that Kim should buy clairvoyance at a price of $15, but what is the value of clairvoyance—the amount she would have to pay for it so that she would be indifferent in buying it and not. The value of clairvoyance is the threshold buying price for clairvoyance.

To compute the value of clairvoyance, we must increase the cost of clairvoyance until she is just indifferent between buying it and not. Thus, in the tree of Figure 3-15, we would consider successively larger values of c like $16, $17, $18, etc. and determine whether it was still a good idea to buy clairvoyance. When we had increased the price to the point where the certain equivalent of the alternative of purchasing clairvoyance was equal to the certain equivalent of not purchasing it, then we would have determined the threshold buying price of clairvoyance. You may wish to verify that at a price of $20 Kim will find that she does not care whether she buys clairvoyance or not because she will face the same $46 certain equivalent regardless of whether she buys it.

We observe that the increase in price from $15 to $20 required to find the value of clairvoyance is just equal to the difference in the certain equivalent of the clairvoyance and no clairvoyance alternatives of Figure 3-15, $51 less $46. While this result is true in our example, it is not generally true for any utility function that a person might have. The only general way to compute the value of clairvoyance is to increase its price iteratively until the point of indifference between buying it and not buying it is reached. We shall soon discuss the special conditions that the utility function must satisfy if we are to be able to compute the value of clairvoyance simply by computing the certain equivalent of the deal with free clairvoyance and then subtracting the certain equivalent of the deal without clairvoyance.

3.1.14.1 Graphical Procedures

We can pursue the issue of valuing clairvoyance graphically as shown in Figure 3-16.

Figure 3-16. Finding the Value of Clairvoyance Graphically

If Kim pays $15 for the clairvoyance, then as we have shown she will face a deal with a 40 percent chance of an $85 payoff and a 60 percent chance of a $35 payoff. To compute the certain equivalent of this deal, we connect the points on the utility function corresponding to $35 and $85 by the dashed straight line. Then we locate a point that is 40 percent of the way up this line and move left to the point C. This is the point that represents the utility of clairvoyance at this price on the u scale, namely 0.674, and the certain equivalent of clairvoyance at this price on the dollar scale, namely $51. Since the point marked I for indoor alternative is otherwise her best choice and this point has a utility of 0.63 and a certain equivalent of $46, the clairvoyant's deal is clearly superior. This merely confirms what we found in Figure 3-15.

To visualize computing the value of clairvoyance from Figure 3-16, we imagine moving the two ends of the dashed line by equal dollar amounts to the left and observing the motion of the point C. We see that as the dashed line moves down the utility function in this way, the point C will slide down until for some total payment to the clairvoyant C will coincide with I. A careful construction will show that this point will be reached when the total payment for clairvoyance is $20.

3.1.14.1.1 Interpretation

The value of clairvoyance can be computed by methods that range from graphical construction to powerful computer programs. Much more important that the method of computation is the interpretation of the results. We must emphasize that the value of computing the value of clairvoyance does not rest in any way on the existence of an entity like the clairvoyant. The point of finding the value of clairvoyance is that it represents the most that the person making the decision should pay for any kind of information on the distinction about which the decision-maker is uncertain. No device, person, survey, or other information gathering process can possibly be worth more than the value of clairvoyance. Knowing the value of clairvoyance, the decision-maker has a benchmark against which to compare any information-gathering scheme that may be proposed. If the cost of the scheme exceeds the value of clairvoyance there is no need to examine the scheme in any further detail. Thus in Kim's case, no source of information about the distinction S/R for weather in the party problem as posed could be worth more than $20 to her.

3.1.15 The Story of Jane

To illustrate how another person might deal with a decision problem about the party, let us consider it from the point of view of Kim's friend Jane. We shall assume that Jane is similar to Kim in many ways. She faces the same party problem, and has the same information about the weather, leading her to assign a probability of 0.4 to the event S. However, Jane's preferences with respect to party prospects may be different from Kim's.

Since Jane has also agreed to follow our five rules of thought, we begin to check her preferences by asking her to order the six prospects for the party. She produces the same ordering as Kim did with an outdoor party in the sunshine O–S being the best prospect and an outdoor party in the rain O–R being the worst. Then, using her agreement to the equivalence rule, we ask her to assign to each of the four intermediate prospects the preference probability for the best prospect versus the worst to which she is indifferent, just as we did for Kim in Figure 3-2. The result is shown in Figure 3-17.

Figure 3-17. Jane's Preference Probabilities

We observe that Jane's preference probabilities are different from Kim's. For example, while Kim said that a porch party in the sunshine was equivalent to a 95 percent preference probability for the best party, Jane says that it is only equivalent to a 90 percent preference probability for the best party. By comparing the two figures we find that Jane's preference probabilities are always less than Kim's for the same prospect, except, of course, for the best and the worst.

We can determine the effect of these different preference probabilities on the choice that Jane would make for party location. We do this by constructing the decision tree of Figure 3-18 and by recording at the endpoint representing each prospect, Jane's preference probability for that prospect as a measure.

Figure 3-18. Jane's Decision Tree

Then, following our usual procedure, we find the preference probability for each of the location alternatives by computing the expected value of this measure for each alternative. For the Outdoor alternative we record $0.4 \times 1 + 0.6 \times 0 = 0.40$; and for the Porch and Indoor alternatives 0.48 and 0.46. Note that the highest preference probability is the 0.48 for the Porch alternative and therefore, by the choice rule, Jane will choose to have her party on the porch. Recall that Kim's best decision was to have the party indoors. Thus the effect of the difference between Jane's and Kim's preference probabilities for prospects is to cause Jane to follow a different alternative for the party location.

Jane, just like Kim, would like to have the advantages of using a monetary measure. Therefore we summon the wizard and investigate Jane's willingness to pay for changes in party prospects. We find that Jane's willingness to pay in every case is exactly equal to Kim's, so the dollar measure she assigns to each prospect is exactly the same. We can record Jane's preference probabilities and corresponding dollar values of each prospect in the table shown in Figure 3-19, just as we did for Kim in Figure 3-9.

Figure 3-19. Jane's Preference Probabilities and Dollar Values

Prospect	Jane's Preference Probability	Jane's Dollar Value
Outdoors, Sunshine	1	\$100
Porch, Sunshine	0.90	\$90
Indoors, Rain	0.50	\$50
Indoors, Sunshine	0.40	\$40
Porch, Rain	0.20	\$20
Outdoors, Rain	0	\$0

We note that Jane's dollar values are directly proportional to her preference probabilities; in fact, they are just the preference probabilities multiplied by 100. This means that if we make a plot of preference probability versus dollar value for Jane, all the points will lie on a straight line. Thus, if we assume that this relationship will apply for any point, then we can plot the utility function for Jane as the straight line shown in Figure 3-20.

Figure 3-20. Kim's and Jane's utility functions

Kim's utility function is also shown for convenience and comparison.

We can use this utility function for Jane to answer the same type of questions we explored using Kim's utility function. For example, to find Jane's certain equivalent for a 50-50 chance at \$100 versus nothing we would proceed as shown in Figure 3-21.

Figure 3-21. Jane's Certain Equivalent for a 50-50 Chance of \$100

We know that for Jane to be indifferent between the deal and the certain equivalent, they must both have the same utility . In the right part of the figure we compute the utility of the deal by first finding the utility of its prospects \$100 and 0 as 1 and 0 by consulting Jane's utility function. Then we find the utility of the deal by finding the expected value of these utilities $0.5 \times 1 + 0.5 \times 0 = 0.5$. The certain equivalent therefore must also have a utility of 0.5. By examining Jane's utility function we find that \$50 has a utility of 0.5. Therefore \$50 is her certain equivalent of a 50-50 chance at \$100.

We observe that Jane's certain equivalent for the deal is in fact equal to the expected value of its monetary value, $0.5 \times 100 + 0.5 \times 0$. We shall see that this is a general property for peop with straight-line utility functions: the certain equivalent of any deal is the expected value of monetary measure.

To illustrate in the party problem, we solve it using monetary values and the utility function just as we did for Kim in Figure 3-14. The solution appears as Figure 3-22.

Figure 3-22. Jane's Certain Equivalents for Party Problem Alternatives

We have recorded both the dollar value and its associated utility for each prospect of the party problem. The utilities for each alternative are, of course, exactly those computed in Figure 3-18. When we compute the certain equivalents of each alternative by finding the dollar amounts corresponding to each utility , we find that they are $40, $48, and $46, for O, P, and I. We observe that these certain equivalents are exactly the expected values of the dollar measures for each alternative. Thus the $48 certain equivalent for the porch alternative equals $0.4 \times 90 + 0.6 \times 20$. We could have found these certain equivalents directly from the monetary measure without ever introducing the particular utility function values.

3.1.16 Attitudes Toward Risk

A person with a utility function that is a straight line will have a certain equivalent for any monetary deal that is the expected value of the monetary measure. We call such a person risk-neutral; Jane is risk-neutral.

We call a person risk averse when he has a certain equivalent for any monetary deal that is always less than or equal to the expected value of the monetary measure. As we saw in Figure 3-14, Kim's certain equivalents for the three party locations Outdoors, Porch and Indoors, $26, $40 and $46 are always less than or equal to Jane's. In fact, if we included more decimal places in our calculation of the certain equivalent for the Indoor alternative, we would see than Kim's certain equivalent for the Indoor alternative is $45.83, slightly less than Jane's, which is exactly $46. Since Kim's certain equivalents are always less than Jane's for the same deals we know that we can call Kim risk averse in this situation. In fact, since her curve as shown in Figure 3-20 is always concave downwards, Kim's certain equivalents will always be less than Jane's unless both are facing certainty.

If a person had a utility function that was always concave upward, then that person's certain equivalents would always be greater than the expected values of the monetary measure. We would such a person risk preferring.

A person who is risk-neutral for the deal under consideration in monetary terms has a considerable practical advantage in being able to compute certain equivalents as expected value of money. As you remember, if we add or subtract a constant from all these measures of a deal, the expected value of the deal changes in exactly the same way. This property is especially useful when we wish to compute the value of clairvoyance for a risk-neutral person. Recall that to compute the value of clairvoyance in general, we had to subtract larger and l costs of clairvoyance from all the prospect measures until the person was indifferent betwe buying the clairvoyance at that price and not buying it. In the case of a risk-neutral person, we know that the certain equivalent for any deal will be its expected value of the monetary measure, and therefore that any subtraction from the measure will cause the same subtraction from the certain equivalent. The practical importance of this is that we can find the value of clairvoyance for risk-neutral persons by computing the expected value of the deal with free clairvoyance and then subtracting the expected value of the best deal without clairvoyance.

Figure 3-23 illustrates this procedure for risk-neutral Jane in the party problem.

Figure 3-23. Determining Jane's Value of Free Clairvoyance on S/R

The upper part of the figure shows Jane's situation without clairvoyance as determined in Figure 3-22. Without clairvoyance she will choose to have the party on the porch and will have a certain equivalent of $48. If she has free clairvoyance, the structure of the tree is exactly like the one for Kim in Figure 3-15. The probabilities and the values are identical; however, there is no subtraction of the $15 cost as there was in the earlier case. We see that if Jane receives the "S" report from the clairvoyant, she will choose to have the party Outdoors and achieve a certain equivalent of $100. If the clairvoyant reports "R", then she will choose to have the party Indoors and will have a certain equivalent of $50. Since the clairvoyant will say "S" with probability 0.4 and "R" with probability 0.6, her certain equivalent with free clairvoyance is $0.4 \times 100 + 0.6 \times 50$ or $70. We know that if she had to pay $70 – $48 or $22 for clairvoyance, then her certain equivalent would be $48 regardless of whether she bought clairvoyance. Therefore Jane's value of clairvoyance on the distinction S/R in the party problem is $22.

As we have seen, it is much simpler to calculate the value of clairvoyance without iteration. We might ask ourselves, is this simplification only available for a risk-neutral person. The answer is no, as we shall soon see. But first we will view the party problem from another perspective.

3.1.17 Mary

Let us consider another friend of both Kim and Jane who shares certain preferences with respect to the party with each of them. Mary has the same preference probabilities for party prospects as does Kim; namely, those shown in Figure 3-9. However, she has the same risk-neutral utility function for money as does Jane in Figure 3-20. This means that her dollar values will be proportional to her preference probabilities as were Jane's in Figure 3-19, but they will be different dollar values from the ones shared by Kim and Jane.

Mary's preference probabilities and dollar values appear in Figure 3-24.

Figure 3-24. Mary's Preference Probabilities and Dollar Values

Prospect	Mary's Preference Probability	Mary's Dollar Value
Outdoors, Sunshine	1	$100

Prospect	Mary's Preference Probability	Mary's Dollar Value
Porch, Sunshine	0.95	$95
Indoors, Rain	0.67	$67
Indoors, Sunshine	0.57	$57
Porch, Rain	0.32	$32
Outdoors, Rain	0	$0

Note that Mary will pay the wizard $67 to move the party from Outdoors to Indoors when rain R occurs, whereas the other girls will pay only $50. All three of them agree that a O–S party is worth $100 more than a O–R party. Although Mary's risk attitude toward money is different from Kim's, she makes exactly the same decision in this situation as does Kim because she has the same preference probabilities. In fact the operations performed for Kim in Figures 3-5 and 3-6 are indistinguishable from those for Mary: both will choose to have the party Indoors. As we can see from these figures, Kim and Mary will make the same decision on party location for any probability of S as long as they agree on that probability. We thus see that we cannot tell whether a person is risk-neutral or risk averse by observing the decision made in any situation that does not have only money as the value measure of each prospect.

When will Mary's actions differ from Kim's? When the clairvoyant arrives. We compute Mary's value of clairvoyance on S using the tree in Figure 3-25 just as we did for Jane in Figure 3-23.

Figure 3-25. Determining Mary's Value of Free Clairvoyance on S/R

In the upper part of the tree we review her decision without clairvoyance. Since she is risk-neutral like Jane, she enters her dollar values for each prospect and computes the expected value of each alternative as its certain equivalent. The Indoor alternative has the highest certain equivalent of $63 ; she will take the action we predicted.

In the lower part of the tree, we compute Mary's value of free clairvoyance. Like Jane and Kim, she will have the party Outdoors when the clairvoyant says "S" and Indoors when he says "R". She assigns, as did Kim and Jane, a $100 value to the report "S"; however, unlike both Kim and Jane, the value she assigns to the report "R" is $67. Her certain equivalent for the party with free clairvoyance is then $0.4 \times 100 + 0.6 \times 67$ or $80. Since Mary is risk-neutral, we can compute her value of clairvoyance on S by subtracting her certain equivalent without clairvoyance from her certain equivalent with free clairvoyance: $80 – $63 = $17.

We have thus found that Kim, Jane, and Mary will each pay different amounts for clairvoyance: $20, $22, and $17, respectively. These differences arise solely from differences in taste, not from differences in structure or information.

4 Risk Attitude

Ronald A. Howard

Section 4: The Party Problem: Risk Attitude

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4.1 Wealth Risk Attitude

When we draw utility functions like those shown in Figure 3-20, showing the dollar values with a range from $0 to $100 is possibly misleading. These dollar amounts are supposed to represent prospects, that is, life with the dollar amount added. This means that zero dollars properly corresponds to the person's present wealth, and any other amount to that amount plus present wealth. If we let w be wealth and w_0 be present wealth, then Figure 4-1 would represent a typical utility function on wealth.

Figure 4-1. A Wealth utility function

Everything we have done up to this point is consistent with the utility function on wealth interpretation. Thus, if Kim's wealth were $13,000 at the beginning of the party problem, we have been using the portion of her utility function on wealth between approximately $13,000 and $13,100 to do our calculations. We invoked this "bank account" idea of wealth in using the wizard to help her assign values.

If a person is given a monetary deal, then we can use the utility function on wealth to calculate the certain equivalent wealth that the person will now have as well as the certain equivalent of the deal. Let the deal x be represented by the possibility of receiving different amounts x_i with corresponding probability p_i. Let w be the person's wealth in general and w_0 be the person's wealth without the deal. Then $\tilde{w}$, the person's certain equivalent wealth with the deal is defined by

$$u(\tilde{w}) = \Sigma_i p_i u(w_0 + x_i)$$

in complete correspondence with our earlier calculations. Since owning the deal is equivalent to having a certain wealth $\tilde{w}$, and since the wealth without the deal is w_0, the certain equivalent of the deal is $\tilde{x} = \tilde{w} – w_0$.

4.1.1 Buying and Selling a Deal

Suppose that a man with wealth w_0 does not own the deal x, but is thinking about buying it. What is his (breakeven) buying price b, the most that he should pay for it? Since when he pays b he is indifferent between buying it at that price and not having it, b is defined by

$$\Sigma_i p_i u(w_0 + x_i – b) = u(w_0)$$

Suppose now that having bought the deal for this price, he wishes to find out how much he should be willing to sell it for, his (breakeven) selling price s. Since he is indifferent between selling it at this price and continuing to own it, s is defined by

$$\Sigma_i p_i u(w_0 + s – b) = u(w_0)$$

The only general solution to this equation is s = b; the amount he is just willing to sell the deal for must be equal to the breakeven amount he paid for it. This is a logical tautology, since he is returning to the state of not having the deal after paying and then receiving the same amount of money.

We obtain the same type of result if the person originally owns the deal in addition to wealth w_0, then sells it and buys it back. We let the (breakeven) selling price in this case be s^*, defined by,

$$u(w_0 + s^* *) = \Sigma_i p_i u(w_0 + x_i)$$

Note that the selling price s^* for an owned deal is just the certain equivalent $\tilde{x}$ for the deal as defined in the preceding section. After selling it for s^*, the breakeven price at which he would buy it back b^* would be defined by

$$u(w_0 + s^*) = \Sigma_i p_i u(w_0 + s^* – b^* + x_i)$$

Again, the only general solution is $s^* = b^*$, equality of the selling price to the subsequent buying price. If a man owns a deal, the price at which he is just willing to sell it must be the price at which he is just willing to buy it back after he sells it.

Notice that these two cycles of purchase and sale take place in different worlds. In both worlds the man has wealth w_0 without the deal. In the first world, the man does not own the deal initially; in the second, he does. Although the buying and selling prices must be the same around each cycle, it does not follow that they are the same in both worlds. It is very possible that $s \neq s^*$ and $b \neq b^*$. The price at which he is just willing to buy something he does not have $b (= s)$ may not be the price at which he is just willing sell the same thing if he has it $s^* (= b^*)$.

We might now ask what would have to be true about a person's preferences if the person were to have $b = s^*$.

4.2 The Delta Property

Consider an additional optional rule that you may wish to be true about your evaluation of monetary deals. Suppose that you own the deal x and that someone offers to increase each and every payoff x_i by the same dollar amount Δ. What do you want to happen to your certain equivalent (selling price) for the deal? For example, Kim's certain equivalent for the 50-50 chance at $0 versus $100 was $34. If now someone sweetens the deal so that she will receive $20 more regardless of what happens, $20 at worst and $120 at best, what should be her certain equivalent for the revised deal? Many would say that her certain equivalent should increase by the same amount that the payoffs have increased, namely, by $20, to a new certain equivalent of $54. The idea is that the $20 is "money in the bank" about which there is no uncertainty.

The delta property is illustrated graphically in Figure 4-2.

Figure 4-2. The Delta Property

On the left side of the figure the person has expressed a certain equivalent $\tilde{x}$ for a three branch deal. On the right side of the figure, the payoffs have changed by Δ and the certain equivalent has changed by the same amount. If this describes the person's wishes for any monetary deal within a range of payoffs, then we say that the person satisfies the delta property within that range.

Within the range where it is valid, the delta property makes the person's certain equivalent for any deal be independent of the person's wealth. The delta property is very attractive to almost everyone for some range of payoffs. However, when Δ is very large and positive, some people say that possession of the deal changes their attitude toward risk so that they are now willing to assess certain equivalents that are much closer to the expected value of payoffs. When Δ is very large and negative others are equally concerned that they will not be sufficiently risk averse.

We shall find it useful to think of the delta property as a convenient approximation to any utility function for a sufficiently small range of payoffs. For some people, this range may in fact be very large. We note that a risk neutral person automatically satisfies the delta property.

4.2.1 Equality of Buying and Selling Prices

Accepting the delta property has strong implications for decision making about monetary deals. To illustrate one of them, consider the three branch deal shown in Figure 4-3.

Figure 4-3. With the Delta Property, Buying and Selling Prices are Equal

On the top left, a woman owns the deal and establishes a selling price $\tilde{x} = s^*$ using the idea that she is indifferent between having s^* for sure and having the payoff from the deal. On the top right, she does not own the deal and establishes a buying price b by finding the amount she would pay for the deal before she would be indifferent to owning it. If she satisfies the delta property, then we know that if we add b to all the payoffs on the right, the resulting deal sho[illegible] on the bottom right must have certain equivalent b, since the certain equivalent must change by the same amount as the payoffs. However, we already know from the top left that this deal has certain equivalent s^*; therefore, $s^* = b$. Thus we have found that if a person satisfies the delta property, her selling price for an owned deal will be the same as her buying price for an unowned deal.

4.2.2 Implications for Computing Value of Clairvoyance

Someone who satisfies the delta property has a much easier task in computing any value of clairvoyance question. Since the certain equivalent for such a person will experience the same change made to all the payoffs of a deal, the person can compute the certain equivalent of a decision situation with clairvoyance when clairvoyance costs k by simply subtracting k from the certain equivalent of the decision situation with free clairvoyance. This means that the value of clairvoyance for such a person is simply the difference between the certain equivalent of the decision situation with free clairvoyance and the certain equivalent of the decision situation with no clairvoyance. There will be no need to use the iterative or graphical techniques for evaluating the value of clairvoyance that we discussed earlier. This simplification is of such great practical importance that it is wise in many cases to assume that the delta property holds exactly when it is close to being acceptable.

4.2.3 Forms of utility function Required by the Delta Property

The delta property puts stringent requirements on the shape of the utility function. If the person has wealth w_0 then the implication of Figure 4-2 is that for any deal and any $\mathit{\Delta}$,

$$u(w_0 + \tilde{x} + \mathit{\Delta}) = \Sigma_i p_i u(w_0 + x_i + \mathit{\Delta})$$

4.2.3.1 Straight Line

Suppose that for any monetary amount y, the utility function has the form of a straight line,

$$u(y) = a + by$$

where a and b are constants, with b positive so that increasing y causes increasing u.

Then,

$$a + b(w_0 + \tilde{x} + \mathit{\Delta}) = \Sigma_i p_i [a + b(w_0 + x_i + \mathit{\Delta})]$$

and since

$$\Sigma_i p_i = 1,$$

we find

$$\tilde{x} = \Sigma_i p_i x_i = E[x|\mathcal{E}].$$

↑

Therefore, as we have noted earlier, the risk neutral person's straight line utility function satisfies the delta property and yields a certain equivalent equal to the expected value of the deal, regardless of Δ and the person's wealth w_0. Jane and Mary satisfied the delta property.

4.2.3.2 Exponential

But are there other forms for the utility function that satisfy the delta property? Yes. By methods that lie outside our scope it can be shown that there is one and only one other form of utility function that satisfies the delta property. It is the exponential form,

$$u(y) = a + bc^y$$

where a, b, and c are constants so defined that increasing y causes increasing u which means, among other things that c cannot be 1.

Substituting this form in the equation for u required by the delta property produces

$$a + bc^{(w_0+\tilde{x}+\Delta)} = \Sigma_i p_i [a + bc^{(w_0+x_i+\Delta)}]$$

and with $\Sigma_i p_i = 1$

$$c^{(w_0+\tilde{x}+\Delta)} = \Sigma_i p_i c^{(w_0+x_i+\Delta)}$$

$$c^{(w_0+\Delta)} c^{\tilde{x}} = c^{(w_0+\Delta)} \Sigma_i p_i c^{x_i}$$

$$c^{\tilde{x}} = \Sigma_i p_i c^{x_i}$$

If the utility function has the exponential form, the certain equivalent of a deal will not depend on Δ or on wealth w_0.

The exponential utility function is described by a single number, c, which may be chosen for convenience. The utility function we used for Kim was of the exponential form,

$$u(x) = \frac{4}{3}\left[1 - \left(\frac{1}{2}\right)^{x/50}\right]$$

Here, $a = 4/3, b = -4/3$, and $c = (1/2)^{(1/50)} = 2^{(-1/50)}$, and we have confirmed that Kim satisfies the delta property.

4.3 Risk Odds

We shall find it useful to provide a direct interpretation for the constant c in the exponential form. For convenience we usually prefer to write the form in terms of the reciprocal of $c, r = c^{(-1)}$, as $u(y) = a + br^{(-y)}$. Suppose that someone satisfying the delta property and, therefore, having a utility function in this form, faces a deal with a probability p of winning one monetary unit and a probability $1 - p$ of losing one monetary unit, as shown in Figure 4-4.

Figure 4-4. Indifference to a Deal Involving Winning or Losing One Monetary Unit

The monetary unit could be any size, like $1000. The question is, for a given r, what p will create indifference between accepting the deal and not accepting it. The figure shows on the right the monetary values and utilities associated with each prospect. The figure shows on the left the monetary and utilities that correspond to the prospect of not accepting the deal. Since for indifference the person must have the same expected value of utilities for not accepting and accepting the deal, we have,

$$a + b = p(a + br^{-1}) + (1–p)(a + br)$$

$$= a + b[pr^{-1} + (1–p)r]$$

or, subtracting a and dividing by b,

$$1 = pr^{-1} + (1–p)r$$

$$(1–p)r^2–r + p = 0$$

Solving for r, we have,

$$r = \frac{1 \pm \sqrt{(1–4p(1–p))}}{2(1–p)}$$

$$= \frac{1 \pm \sqrt{(1–4p + 4p^2)}}{2(1–p)}$$

$$= \frac{1 \pm (1–2p)}{2(1–p)}$$

or,

$$r = 1, \frac{p}{(1–p)}$$

The solution $r = 1$ is a limiting case that corresponds to the risk-neutral utility function we have discussed previously, $u(y) = a + by$. The solution $r = p/(1–p)$ means that when r is equal to the odds of winning one monetary unit versus losing one monetary unit, the person is indifferent between accepting and rejecting the deal. We call r the risk odds that characterize the risk attitude of a person who satisfies the delta property.

4.3.1 Using Risk Odds

The constant r can be put to immediate use. Suppose that the person faces a deal of winning or losing not just one, but m monetary units as shown in Figure 4-5.

Figure 4-5. Indifference to a Deal Involving Winning or Losing m Monetary Unit

What is the probability of winning q that will just create indifference? We proceed as before,

$$a + b = q(a + br^{-m}) + (1-q)(a + br^{m})$$
$$= a + b[qr^{-m} + (1-q)r^{m}]$$

then,

$$1 = qr^{-m} + (1-q)r^{m}$$
$$(1-q)r^{2m} - r^{m} + q = 0$$

By comparing this equation with our earlier development, we see that the solution for r to the power m must be $r^m = q/(1-q)$, or since $r = p/(1-p)$,

$$\frac{q}{1-q} = \left(\frac{p}{1-p}\right)^m$$

The odds required to be indifferent for stakes m are just the odds required to be indifferent for unit stakes raised to the mth power. This is true for any value of m for an exponential person. For example, if the stakes were one-half a monetary unit, then the odds on winning for indifference would be just the square root of the risk odds.

4.3.2 Kim's Risk Odds

Let us test these ideas in Kim's case. We have already found that for Kim,

$$c = (1/2)^{(1/50)} = 0.986$$

and hence that her risk odds are

$$r = 1/c = (1/2)^{(-1/50)} = 1.014$$

This means that Kim would have to have 1.014 odds of winning versus losing $1 before she would be indifferent.

More interesting, however, are the odds of winning versus losing $100 that would just make her indifferent. These odds would be just the 100th power of her risk odds or

$$[(1/2)^{(-1/50)}]^{100} = (1/2)^{-2} = 4$$

Kim would require 4:1 odds of winning versus losing $100 to be indifferent. This means that she would have to have a 0.8 chance of winning and 0.2 chance of losing.

4.3.3 Assessing Risk Odds

We have now found one way we could have assessed Kim's utility function. First we would ask questions that confirmed that she wanted to satisfy the delta property for gains and losses in the $100 range. Then we would ask her what chance of winning versus losing $100 would create indifference. When she says 80 percent or 4:1 odds, we can use this answer to construct her utility function and answer any question regarding risky choice that she might face in the specified range. From our discussion, it follows that Kim is just indifferent to 2:1 odds of winning versus losing $50.

We can use the same method to assess the utility function for anyone who accepts the delta property. The best way is to select a large enough monetary unit like $100 or $1000 to gain the person's interest. Then you can ask the odds for winning versus losing that would create indifference. You can check for consistency by doubling or halving the size of the monetary unit and repeating the question. If the person is always indifferent to equal odds, then we are observing risk neutrality and we use the utility function form $u(y) = a + by$.

4.3.4 Relationship to Other Forms of the Exponential

We sometimes prefer to write the exponential in other forms than the one we have used here. If we let $r = e^{\gamma}$ and therefore $\gamma = \ln r$, we have defined the risk aversion coefficient γ. Its reciprocal ρ, the risk tolerance, is given by $\rho = 1/\gamma = 1/\ln r$. Both γ and ρ are useful in various practical and theoretical discussions.

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5 Sensitivity Analysis

Ronald A. Howard

Section 5: The Party Problem: Sensitivity Analysis

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5.1 Sensitivity Analysis

Decision analysis can create important insights into the nature of a decision by allowing us to perform a sensitivity analysis on the numbers that constitute the decision basis. By seeing how the decision would change if these numbers were changed, we can determine whether additional effort should be expended in increasing the precision of the numbers we have used. Sensitivity analysis is an extremely important feature of professional decision analysis, where we must continually refine and focus our attention on the important aspects of the problem. In future sections we shall discuss the subject of sensitivity analysis in much more detail. However, let us now illustrate its use in the Party Problem.

5.1.1 Kim's Sensitivity to Probability of Sunshine

Suppose that Kim is concerned about how sensitive her decision is to the probability of sunshine she assigns; she has currently assigned 0.4. She may want to know whether small changes in this probability will cause changes of alternative and/or substantial changes in the value of the party to her. She may also be interested in a sensitivity analysis to this probability, because she expects to receive additional information and wants to know how she should adapt her party strategy in the face of a new probability of sunshine.

To explore this issue, we let p be the probability of sunshine that she assigns and consider how the analysis of the problem will depend on p. In Figure 5-1 we draw Kim's decision tree for a probability of sunshine p replacing the value of 0.4 we used in Figure 3-14.

Figure 5-1. Kim's Decision Tree for General Probability of Sunshine, p

The utilities of the three alternatives are easily computed. The utility of the outdoor alternative is simply p. The utility of the porch alternative is $0.950p + 0.323(1-p)$ or $0.323 + 0.627p$. The utility of the indoor alternative is $0.568p + 0.667(1-p)$ or $0.667 - 0.099p$. The utility for each alternative is shown by the branch for that alternative in the figure.

Figure 5-2 shows a plot of these utilities versus p, the probability of sunshine, as p ranges from 0 to 1.

Figure 5-2. Kim's utility Sensitivity to Probability of Sunshine

Each alternative plots as a straight line that joins the utility of the alternative with $p = 0$ to the utility of the alternative with $p = 1$. The best alternative to follow is the one with the highest utility for the particular p. Thus, we see that the best alternative when $p = 0$ is the indoor alternative. The indoor line remains highest until it crosses the porch line at a value of $p = 0.47$. Then the porch line is highest until it crosses the outdoor line at $p = 0.87$. Thus, for probabilities of sunshine less than 0.47, the indoor alternative is best; for probabilities of sunshine greater than 0.87, the outdoor alternative is best; and otherwise the porch alternative is best. Note that the value of probability of sunshine that Kim originally assigned, 0.4, leads her to follow the indoor alternative, as we found in Figure 3-14. You can see that Kim will only move to the outdoor alternative when is very sure that the weather will be sunny.

One easy way to draw Figure 5-2 is to create on the right hand side of the figure a dollar scale that is distorted according to Kim's utility function of Figure 3-11. For example, the point corresponding to $50 on the right of Figure 5-2 is found by observing from Figure 3-11 that $50 corresponds to a utility of 0.667. Once we have constructed this distorted dollar scale, then we can draw the lines corresponding to each alternative by simply connecting the end points th correspond to the dollar values. Thus, for example, the porch line connects the point corresponding to $20 when $p = 0$ to the point corresponding to $90 when $p = 1$.

We can also use this figure to show the utility of free clairvoyance. Recall that if Kim receives free clairvoyance on the weather and, if she finds that the weather will be sunny, then she will have a party outdoors with a value of $100. If she finds that the weather will be rainy, then she will have her party indoors with a value of $50. Consequently, if the probability of sunshine is p, the utility of free clairvoyance is $pu(100) + (1 - p)u(50)$. This is the line that connects the point corresponding to $50 when $p = 0$ to the point corresponding to $100 when $p = 1$. It is shown dotted in Figure 5-2. Note that it connects the highest point on the $p = 0$ axis to the highest point on the $p = 1$ axis. It is clear that the alternative consisting of free clairvoyance and then a decision is better than any of the other alternatives in the problem without clairvoyance unless $p = 0$ or $p = 1$.

5.1.1.1 Certain Equivalent Sensitivity

For any alternative, and any probability of sunshine p, we can determine from Figure 5-2 the corresponding utility and, consequently, by reading off the distorted dollar scale, the corresponding certain equivalent of that situation. However, it will be more convenient to perform this operation on the whole figure and so obtain a plot of Kim's certain equivalent sensitivity to probability of sunshine p as shown in Figure 5-3.

Figure 5-3. Kim's Certain Equivalent Sensitivity to Probability of Sunshine, p

Note that these plots showing how the certain equivalent of each alternative depends on the probability of sunshine are not straight lines because of the distortion in the dollar scale of Figure 5-2. We see, for example, that with Kim's original probability of sunshine $p = 0.4$, the indoor, porch, and outdoor alternatives have, respectively, certain equivalents of $45.83, $40.60, and $25.73, in agreement with Figure 3-14. We also see that at this point her certain equivalent for free clairvoyance is $66.10. Since Kim satisfies the delta property, this means that the value to her of free clairvoyance is 66.10-45.83 or $20.27, in agreement with our earlier results.

5.1.1.2 Value of Clairvoyance Sensitivity

Because Kim wishes to accept the delta property, we can in fact find her value of clairvoyance for any probability of sunshine p as the difference in heights in Figure 5-3 between the certain equivalent for free clairvoyance and the certain equivalent for the best alternative. The result of the computation appears as Figure 5-4.

Figure 5-4. Kim's Value of Clairvoyance Sensitivity to Probability of Sunshine, p

We observe that there will be no value of clairvoyance when p is either 0 or 1 because at these points Kim would have clairvoyance. Otherwise, in this problem, there will be a positive value to free knowledge of the weather. As p increases from 0, the value of clairvoyance also increases, reaching a peak of about $24.36 at $p = 0.47$. Note that the value is $20.27 at $p = 0.4$. When p

exceeds 0.47, the value of clairvoyance falls, and falls even more rapidly when p exceeds 0.87. We can see that clairvoyance will be worth at least $15 for probabilities of sunshine that range from about 0.3 to 0.85. Figure 5-3 also shows that as the probability of sunshine varies from 0.2 to 0.6. Kim's certain equivalent will be much less sensitive to this change if she follows the indoor alternative than it would be if she followed the porch alternative. There is only about a $4 difference in certain equivalents over this range for the indoor alternative, while there would be a $24 difference for the porch alternative.

5.1.2 Jane's Sensitivity to Probability of Sunshine

We can perform this analysis even more easily for the risk-neutral Jane. Figure 5-5 shows Jane's decision tree for the general probability of sunshine p in correspondence with Figure 3-22.

Figure 5-5. Jane's Decision Tree for General Probability of Sunshine, p

Since Jane's certain equivalents are simply her expected value of monetary values, we find her certain equivalent for the outdoor alternative is simply $100p$. For the porch alternative, her certain equivalent is $90p + 20(1-p) = 20 + 70p$; for the indoor alternative, it is $40p + 50(1-p) = 50 - 10p$. We plot these certain equivalents in Figure 5-6.

Figure 5-6. Jane's Certain Equivalent Sensitivity to Probability of Sunshine, p

The certain equivalent for each alternative is a straight line connecting the dollar values for $p = 0$ and $p = 1$. Again, as p increases from 0 through 1, the best alternative will change from indoor to porch to outdoor; however, the points of change are different from those for Kim. Jane will change from the indoor to the porch alternative when p crosses 0.375 and from the porch to the outdoor alternative when p crosses 0.667. Thus, she is more willing to follow the more exposed alternatives for lower probabilities of sunshine than is Kim. We see that the value of $p = 0.4$ corresponds to the porch region for Jane. Her certain equivalents are shown for the indoor, porch, and outdoor alternatives for $p = 0.4$ as $48, $46, and $40.

If Jane is given free clairvoyance, her certain equivalent is represented by the dotted line that connects the $100 value she will receive when $p = 1$ by having the party outdoors and the $50 value she will receive when $p = 0$ by having the party indoors. When $p = 0.4$ this line shows that Jane's certain equivalent of free clairvoyance will be $70. Since Jane is risk-neutral, she satisfies the delta property and consequently we can obtain her value of clairvoyance at $p = 0.4$ by subtracting from the $70 certain equivalent of free clairvoyance the $48 certain equivalent of her otherwise best alternative, having the party on the porch. The result is the $22 value of clairvoyance for Jane that we computed in Figure 3-23.

We can perform this same computation of subtracting the certain equivalent of the best alternative for a particular p from the certain equivalent of free clairvoyance at the same p to obtain Jane's value of clairvoyance sensitivity to p as shown in Figure 5-7.

Figure 5-7. Jane's Value of Clairvoyance Sensitivity to Probability of Sunshine, p

Just as for Kim, Jane will have no value of clairvoyance when $p = 0$ or $p = 1$ because she will already have clairvoyance at these points. Otherwise, clairvoyance will have a positive value to Jane. As you can see from the construction, the segments of Jane's value of clairvoyance sensitivity curve will all be straight lines. The highest value of clairvoyance occurs at the crossover point $p = 0.375$ and here it is $22.50. For the probability of sunshine 0.4, the value of clairvoyance is $22 as we have observed.

5.1.3 Comparison of Kim's and Jane's Value of Clairvoyance Sensitivities

We can now compare the value of clairvoyance sensitivity for both Kim and Jane. Note that, from Figure 5-4, when the probability of sunshine is 0.4, Kim's value of clairvoyance is $20.27, whereas from Figure 5-7, Jane's value of clairvoyance is $22. Kim, as the risk averse person, is in this case willing to pay less for clairvoyance than is Jane. But is that always true? While many people think so, the answer is no. Compare the two figures when $p = 0.5$. At this point Kim's value of clairvoyance is about $24, while Jane's is $20. The order of their values is reversed for this different probability of sunshine.

One way to interpret this result is to note that when $p = 0.5$, both Kim and Jane will follow the porch alternative, and be more exposed to the weather. Kim, as a risk averse person, is more concerned about this and consequently is willing to pay more for clairvoyance on the weather. However, when $p = 0.4$, Kim has already switched to the indoor alternative, while Jane is still following the porch alternative. Kim, as we found in the sensitivity analysis, is now much less exposed to the weather and so her value of clairvoyance becomes less than Jane's. Even in such a simple problem as this, intuition can often be misleading.

By superimposing the value of clairvoyance sensitivities for both Kim and Jane as shown in Figure 5-8, we observe that these two people who share everything but risk attitude will in fact have quite different dependence of value of clairvoyance on probability of sunshine p.

Figure 5-8. Value of Clairvoyance Sensitivity to p for Kim and Jane

We see that Jane will be willing to pay more for clairvoyance for values of p below 0.423, and that Kim will be willing to pay more for clairvoyance for values of p above 0.423.

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6 Basic Information Gathering

Ronald A. Howard

Section 6: The Party Problem: Basic Information Gathering

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6.1 Basic Information Gathering

6.1.1 The Value of Experimentation

One of the most useful features of decision analysis is its ability to distinguish between constructive and wasteful information gathering. A relevance diagram for a typical information gathering or experimental process is shown in the upper part of Figure 6-1.

Figure 6-1. Relevance Diagrams for Experimentation

Here we have a distinction of interest and another related observed distinction that will be the result of the experimental process. The value of the observed distinction, in fact, derives from the relevance between it and the distinction of interest that is indicated by the relevance arrow. The distinction of interest could be the amount of oil in a particular underground structure, the observed distinction could be the result of seismic tests. The distinction of interest could be future product sales, the observed distinction could be the results of a market survey. Or we could be dealing with a disease and the results of a medical test; in-service manufacturing costs, and pilot plant manufacturing costs; or any of the myriad other information gathering activities that form a support for decision making.

Notice that which distinction we regard as the distinction of interest and which as the observed distinction is somewhat arbitrary. Using our earlier example, we could consider the extent of beer drinking as the result of a test that is relevant to the amount of education, or the amount of education as a test that is relevant to the extent of beer drinking. The context of the decision problem will usually indicate which of the uncertainties is to be regarded as the distinction of interest and which as the observed distinction.

6.1.1.1 Distinction of Interest

The distinction of interest could be the amount of oil in a particular underground structure, the observed distinction could be the result of seismic tests. The distinction of interest could be future product sales, the observed distinction could be the results of a market survey. Or we could be dealing with a disease and the results of a medical test; in-service manufacturing costs, and pilot plant manufacturing costs; or any of the myriad other information gathering activities that form a support for decision making.

6.1.1.2 Observed Distinction

Notice that which distinction we regard as the distinction of interest and which as the observed distinction is somewhat arbitrary. Using our earlier example, we could consider the extent of beer drinking as the result of a test that is relevant to the amount of education, or the amount of education as a test that is relevant to the extent of beer drinking. The context of the decision problem will usually indicate which of the uncertainties is to be regarded as the distinction of interest and which as the observed distinction.

6.1.2 Assessed Form

The upper relevance diagram in Figure 6-1 is called the assessed form because it is the form in which probabilities representing our information are usually assessed. The arrow from the distinction of interest to the observed distinction represents the conditional probability of the observed distinction given the distinction of interest. This is often called the "likelihood." The other probability assessment needed to complete this form is the probability distribution on the distinction of interest. It is usually called the "prior" because it is usually assessed before the results from the experimental process are known.

6.1.3 Inferential Form

However, the information contained in this relevance diagram must be placed in the inferential form shown in the lower part of Figure 6-1 if we are to obtain information from the experimental process in its most useful form. Here the arrow from observed distinction to distinction of interest shows us the probability distribution that we will have to assign to the distinction of interest when the observed distinction is known. This is usually called a "posterior" distribution because it is useful after the results of the experiment are known. The other probability distribution implied by the inferential form is the probability distribution on what will be observed in the experiment—the probability distribution on the observed distinction. This is the "preposterior" distribution. Note that if there is no uncertainty in what will be observed, there is no point in performing the experiment.

It is clear from our earlier work that transforming the assessed form into the inferential form is just a matter of the arrow reversal or tree reversal that we discussed in our section on characterization. Every information gathering process, regardless of its form, logically requires the steps that we have described here. First, the description of the experiment in terms of the likelihood and the description of our original knowledge in terms of the prior. Then this information must be processed to provide both the posterior distribution, which shows wha
will learn from different results of the experiment, and the preposterior distribution, which wi

show us the chance of obtaining each of these results. The nature of this entire experimental process will be made most clear by applying it in the Party Problem.

6.1.4 The Acme Rain Detector

Suppose that, as Kim is preparing to decide on her party location, she is approached by someone who offers her the services of an Acme Rain Detector. The detector will provide either an indication of sunshine "S" or of rain "R." Extensive testing of the detector's operation has shown that when the weather will actually be sunny, there is an 80 percent chance that the detector will indicate sunshine and, that when the weather will actually be rainy, there is an 80 percent chance that the detector will indicate rain. Thus, the detector's indication is a useful but not infallible prognostication of the weather.

The experimental relevance diagrams for the Acme Rain Detector appear in Figure 6-2.

Figure 6-2. Relevance Diagrams for the Acme Rain Detector

The actual weather, that is, sunshine S or rain R, is definitely relevant to the detector's indication. This is the diagram that we must assess. However, when it is assessed, then we will find that we are using it in the form shown in the lower part of the figure where we will be seeking to assign a probability to the weather given the detector's indication.

The person offering the rain detector service, however, is not an altruist. He seeks a fee of $10 for the detector's services. So, Kim's decision has now expanded to whether she should pay $10 to use the detector. Before we proceed any further, Kim should review the implications of her value of clairvoyance on the weather. We found that she would be willing to pay as much as $20 (in fact, $20.27) to have perfect information on the weather. If the Acme service had requested more than this amount, then she would have to do no further analysis before rejecting it. However, its price is about half her value of clairvoyance, so there is at least a possibility that using it may be a good idea.

6.1.4.1 Assessed Form

To be sure, she will have to do some further analysis. She begins by representing the likelihood for the detector in tree form as shown in Figure 6-3.

*Figure 6-3. Assessment of Detector Properties^*

We see that when the weather will in fact be sunny there is an 80 percent chance of a detector indication of sunshine; when the weather will be rainy, then there is an 80 percent chance that the detector will indicate rain. Kim must supplement this with her prior probabilities of the weather—0.4 for sunshine and 0.6 for rain—in order to have a complete description of the experimental situation. This appears in tree form in Figure 6-4.

Figure 6-4. Complete Assessment of Experimental Situation

We know that in order to use the information this tree contains we will have to reverse its order. As a first step, we compute the probabilities of each of the four elemental possibilities implied by the tree by multiplying together the probabilities along each path leading to each elemental possibility.

6.1.4.2 Inferential Form

In Figure 6-5 we have reversed the tree order, using our customary procedure.

Figure 6-5. Experimental Inference

We have copied the probabilities of the elemental possibilities to the correct end point of the reversed tree and then used our standard rules to distribute probabilities among the branches. We see, for example, that there is a 0.44 chance that the detector will indicate sunny, "S", before it is turned on, a result obtained from adding together 0.32 and 0.12. This is the preposterior probability because it indicates the chance of the experimental indication before the experiment is performed.

The posterior probabilities on the weather, given the experimental indication, are those shown at the second layer of the tree. Since the odds are 32 to 12 that it will be sunny when the detector says sunny or "S", there is a probability 8/11 = 0.727 of sunshine following such an indication. However, if the detector indicates rainy or "R" then there are 8 to 48 odds or probability 1/7 = 0.143 of sunshine. Note that the one thing Kim will be sure of is that after she turns on the detector, her probability of sunshine will no longer be 0.4. It will become either 0.727 if the detector indicates sunny or 0.143 if the detector indicates rainy. The chance of each of these possibilities is 0.44 and 0.56 respectively.

6.1.5 Detector Use Analysis for Kim

Now we are ready to find whether Kim should pay $10 to use the detector. We construct a decision tree similar to that of Figure 3-15; in fact, since the upper part of the figure where she makes the decision without any further information is identical to that of Figure 3-15, we represent it here only in summary form. Recall that in this case she will have the party indoors and will have a certain equivalent of $46, or $45.83 to be more precise. However, if she buys the rain detector for $10, then she faces the tree shown in the lower part of Figure 6-6.

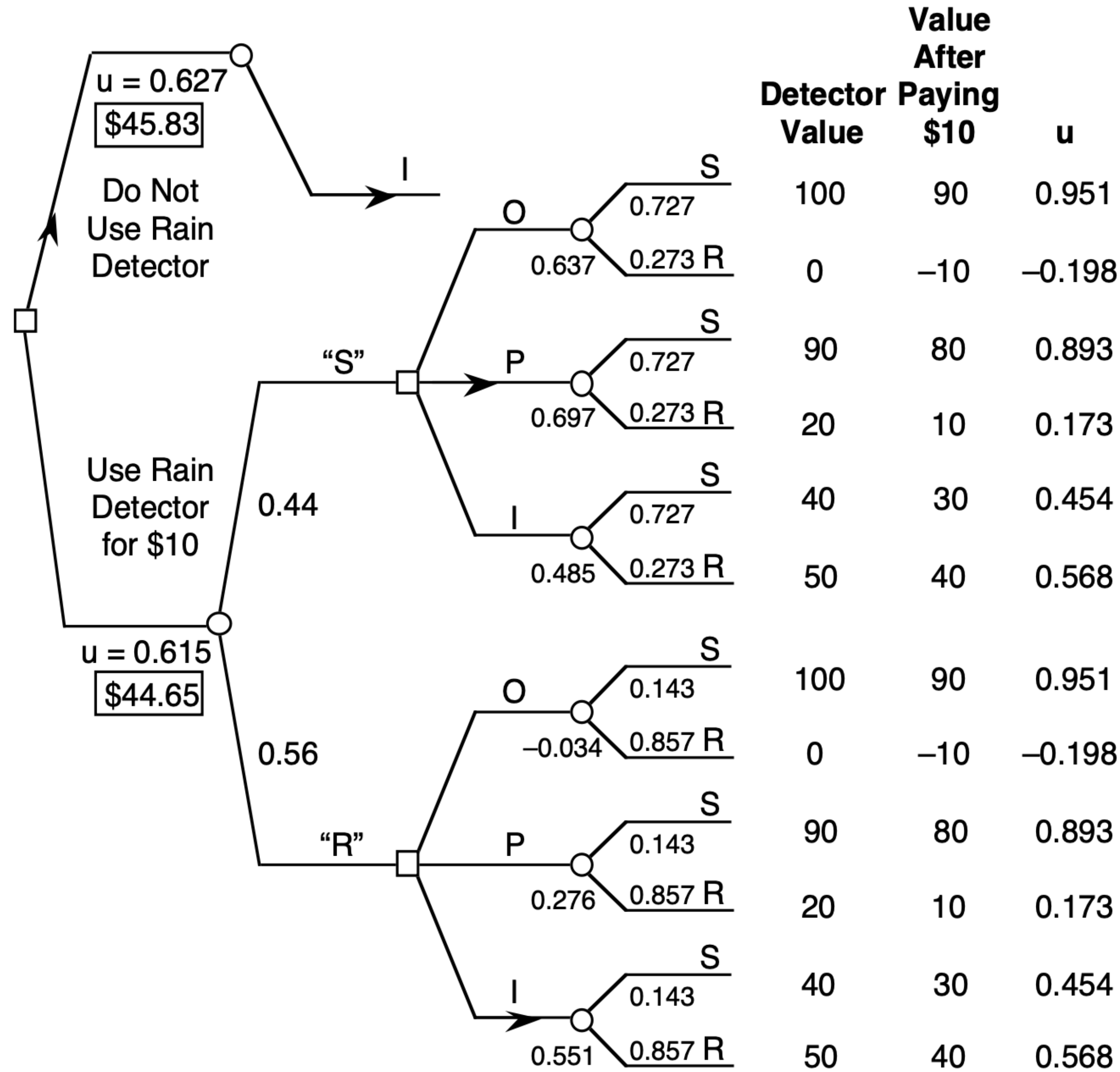

Figure 6-6. Should Kim Pay $10 to Use the Rain Detector

We shall analyze the problem without using our knowledge that Kim satisfies the delta property so that we will be able to observe a general analysis. Note that the structure of the problem is exactly the same as that for clairvoyance, the only difference is in the probabilities that we will be assigning.

The values at the tips of the tree are the usual dollar values. We begin by subtracting the $10 cost of the experiment from each of the values and then computing the utility of each of these reduced amounts, as shown in the column at the far right of the figure. Then we copy the appropriate probability assignments into the figure, for example, the preposterior probabilities of 0.44 and 0.56 that the detector will indicate sunny and rainy, respectively. Next, following each indication, we record the proper conditional probability of sunshine and rain as computed in

Figure 6-5. Thus, following the sunny indication, there will be a 0.727 chance of sunshine and, following the rainy indication, there will be a 0.143 chance of sunshine. Now we can roll back this tree to see whether the utility of buying the rain detector service is greater than the utility of 0.63 of doing no experiment or, equivalently, whether the certain equivalent of buying the rain detector at a cost of $10 is greater or less than $45.83, the certain equivalent of performing no experiment.

Proceeding, we find that following the branch of a sunny indication, "S", the decision to have the party outdoors will produce a 0.727 chance of a 0.951 utility and 0.273 chance of a -0.198 utility , or a utility for this alternative of 0.637. The porch alternative in this situation produces a utility of 0.697, and the indoor alternative a utility of 0.485. Consequently, the porch alternative is best and, if the detector indicates sunny, she will decide to have the party on the porch. Notice that this is a change from the decision she would make otherwise to have it indoors.

The alternatives following the rainy indication have utilities -0.034, 0.276, and 0.551. Clearly the indoor alternative is best when the detector indicates rainy. This is what she would have done if she did not use the detector, so the detector has the possibility of changing her decision only when it indicates sunshine. Therefore, overall she has a 0.44 chance of a utility of 0.697 and a 0.56 chance of a utility of 0.551, producing a utility of 0.615 with a corresponding certain equivalent of $44.62 for the rain detector alternative. We see that Kim is better advised to pass up the opportunity of using the rain detector because she will be worse off by at least $1 by doing so. Thus, even though the detector has an 80 percent chance of giving a correct indication, and even though her value of clairvoyance is at least $20, Kim would not be wise to employ the detector in this situation.

6.1.5.1 Analysis Using the Delta Property

As we mentioned, we have ignored the fact that Kim satisfies the delta property and carried out the analysis in some detail to show the general procedure for dealing with experimentation. We have also used more accuracy than necessary to allow following the computations in detail. However, let us now take advantage of Kim's acceptance of the delta property. We do this by noting that we have already solved the problem of what Kim would do for any probability of sunshine that she might face in Figure 5-2. If Kim faced a 0.727 chance of sunshine, we see that she would follow the porch alternative and if she faced a 0.143 chance of sunshine, we see that she would follow the indoor alternative. Therefore, assuming that she received use of the rain detector at no charge, her utility of using the free detector would be the chance it would indicate sunny, 0.44, times the utility of the porch alternative when the probability of sunshine is 0.727, or $0.323+0.628(0.727)=0.780$ plus the chance that the detector will indicate rainy, 0.56, times the utility of the indoor alternative when the probability of rain is 0.143, or $0.667-0.099(0.143)=0.653$. The result is a utility of 0.709 or a certain equivalent of $54.7 means that Kim would have a certain equivalent of about $54.7 for the experiment if she co get it free. Since her certain equivalent without the experiment is about $46.00, she should pay no more than about $8.7 for the experiment as a believer in a delta property. The cost of $10 is over a dollar too high.

6.1.6 Detector Use Analysis for Jane

We can use this same approach to determine rather quickly whether the detector is a good idea for Jane. Recall that Jane's value of clairvoyance was $22.00. We use Figure 5-6 to compute Jane's certain equivalents for each possible detector reading. If the detector reads "S", then the probability of sunshine will be 0.727 and she will choose to have the party outdoors with a certain equivalent of 100p = $72.70. If the detector indicates rain "R", then the probability of sunshine will be 0.143 and she will choose to have the party indoors with a certain equivalent of $50 - 10p$ or $50 - 10(0.143) = 48.57$. Using the probabilities of these indications, we obtain the certain equivalent of a free detector to the risk-neutral Jane as 0.44(72.70)+0.56(48.57) = $59.19. Since Jane's certain equivalent of the party without using the detector is $48.00 obtained by having the party on the porch, the value of the detector to her is $11.19. If the detector is offered at a cost of $10.00, Jane will make a profit of over $1.00 by using it. Notice that, in Jane's case, Jane's decision will change from her current alternative of having the party on the porch to either having the party outdoors for a sunshine indication or indoors for a rain indication. Therefore, Jane will benefit from both indications of the detector, while Kim will benefit from only one. As a result, using the Acme Rain Detector at a cost of $10 is a good deal for Jane but a bad deal for Kim.

6.1.7 General Observations on Experimentation

Experimental decisions, no matter how complicated, can be handled by exactly the same principles we have used in discussing experimentation in the Party Problem and with exactly the same tools. We should note that an experiment cannot have value unless there is the possibility that it will change the alternative used by the decision-maker.

There are several principles that guide experimentation or information gathering. First, an information gathering process is worth considering only if its results are relevant to distinctions of interest in the decision problem. Second, an information gathering process is worth considering only if the observations it produces have the possibility of changing the decision that is made. We say that the process must be material to the decision. Third, an information gathering process is worth considering only if using it is economic in terms of the overall decision. The Acme Rain Detector was relevant and material for both Kim and Jane, but economic only for Jane. The idea of the value of an experiment already resides in the value of clairvoyance, for the value of an experiment is the value of clairvoyance on the observations of the experiment.

6.1.8 Decision Diagrams

The concept of relevance diagrams, so useful in understanding uncertainty, can be extended to provide comparable insights into decisions. We first add a new kind of node called a decision node to represent the selection of an alternative; it is represented by a square or rectangular box. Arrows entering decision nodes specify that the decision is made with the knowledge of the quantity at the other end of the arrow; they are informational arrows.

We also introduce a value node to represent the value measure that the decision-maker is seeking to maximize. The value node is usually shown as an octagon. Arrows entering value nodes show the quantities on which the value depends, whether those quantities be decisions or other distinctions. There can be no uncertainty in a value node once the quantities that enter it are specified; we say that a value node is deterministic.

The simplest decision diagram consists of a single decision node and a single value node, as shown in Figure 6-7.

Figure 6-7. The Simplest Decision Diagram

Such a diagram would imply a decision process consisting of selecting the alternative with the highest corresponding value.

Most interesting decision problems have at least one uncertain node and possibly more than one decision node. For example, in our original Party Problem, Kim must make a decision about party location when she is uncertain about the weather. Her value for the party depends on the location: outdoors, porch, or indoors, and on the weather, sunshine or rain. The corresponding decision diagram for Kim's problem appears as Figure 6-8.

Figure 6-8. Decision Diagram for Party Problem

Notice that there is no arrow entering the weather node because none of the other quantities we have discussed is relevant to the weather.

Note also that this decision diagram for the problem could have been drawn before we made any judgments about the number of alternatives to be considered for location or the number of degrees of distinction to be used for the weather. We could draw it without specifying probability assignments on weather or particular value numbers to be associated with combinations of location and weather. In fact, drawing a diagram in this form is the most convenient place to begin a discussion of the problem. Once we can agree on its basic structure, as in Figure 6-8, then we can proceed to the more detailed analysis that we have in fact carried out. Figure 3-14 shows the specific decision tree that we constructed to represent the decision diagram of Figure 6-8.

6.1.9 Value of Clairvoyance

As we have said, an arrow into a decision node means that the information at the other end of the arrow is known at the time of the decision. If we wish to consider the value of clairvoyance on weather, for example, all we must do is construct the decision diagram of Figure 6-9 where an arrow connects weather to location.

Figure 6-9. Decision Diagram for Value of Clairvoyance on Weather

The certain equivalent of the best alternative in this decision diagram would represent the certain equivalent of the deal with free clairvoyance on the weather. If Kim satisfies the delta property, as she does, then we would only have to subtract from this quantity the certain equivalent of the best deal without clairvoyance on the weather, as computed corresponding the Figure 6-8, to find the value of clairvoyance on weather to Kim.

6.1.10 Experimental Information

If Kim did not receive free clairvoyance on the weather, but rather a free Acme Rain Detector service, then her decision diagram would appear as in Figure 6-10.

Figure 6-10. Decision Diagram with Free Detector in Assessed Form

Here she will know the detector indication when she makes the location decision. Note that there is a relevance arrow connecting weather to detector indication to reflect the assessment required in the upper part of Figure 6-2.

6.1.10.1 Decision Tree Order

We know that to use this information in tree form, we must reverse the arrow between weather and detector indication as shown in the bottom of Figure 6-2. This change in the decision diagram if reflected in Figure 6-11.

Figure 6-11. Decision Diagram with Free Detector in Decision Tree Form

Recall that in a decision tree, the convention is that all distinctions that precede a decision node are known at the time a decision is made. Note that this is true of Figure 6-11 but not of Figure 6-10. If we attempt to draw a decision tree from Figure 6-10, we would have to start out with weather, then detector indication, then location, and then value. However, this would not be correct because the tree structure would imply that weather is known when location is chosen and that is not so with Figure 6-10. However, in Figure 6-11, the order would be detector indication, location, weather, and value. The tree can be drawn in this order and its certain equivalent would be Kim's certain equivalent for free use of the detector.

We can describe what is necessary to place a decision diagram in decision tree order quite succinctly by thinking of each arrow in a diagram as representing a parent to child relations using conventional genealogical terms. Thus, in Figure 6-10 we can say that the location node has two ancestors—detector indication and weather. However, it has only one parent—detector indication. To place a decision diagram in decision tree order form we must reverse relevance arrows until the only ancestors of any decision node are its parents. Notice that this is the case for Figure 6-11.

6.1.11 Detector Use Decision

If we wish to help Kim determine whether she should use the detector at a specified price, we might begin by drawing a decision diagram like that in Figure 6-12.

Figure 6-12. Possibly Confusing Decision Diagram for Party Problem with Detector

We have now added a use detector decision node with three arrows emanating from it. One arrow goes to the value node to show the effect on value of the cost of using the detector. A second arrow goes to location to show that, when Kim makes the location decision, she will know whether she used the detector. We presume in this, as in all decision diagrams, that decisions previously made are never forgotten at the time of future decisions. Finally, an arrow from the use detector node enters the detector indication node in an attempt to show that the indication observed will depend on whether or not we use the detector.

While this diagram can be used with care, it can create a possible confusion because the arrow between the use detector node and the detector indication node makes it look as if the actual indication of the detector may be changed by whether or not it is used, and that is not our intent.

6.1.11.1 Deterministic Nodes

A more accurate decision diagram to represent the possibility of purchasing the use of the detector is shown in Figure 6-13.

Figure 6-13. More Accurate Decision Diagram for Party Problem with Detector

Here we have added a new node called "detector report" in place of the original detector indication and have shown the detector indication as an input to the detector report node. This detector report node has a double circular border to indicate that it is a deterministic node: that its outputs are known when its inputs are known. This means that the detector report that will be available at the time the location decision is made will depend on both the decision on using the detector and on the detector indication that would be obtained if it were used. One possible detector report is no indication as the result of a decision not to use. This decision diagram would lead directly to the decision tree of Figure 6-6 for the actual determination of whether Kim should use the rain detector for a cost of, for example, $10.

As we have seen, one or more relevance diagrams may form a major portion of decision diagrams. Everything that we have learned about the manipulation of relevance diagrams is directly relevant when they appear in decision diagrams.